

Remember What You Thought, Not What You Said: Self-Distilled Memory Admission for LLM Agents

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Abstract

LLM agents must decide what to remember before a future question arrives. Useful concepts are often inferences the passage never states. Importance judgments rank candidates by provenance, favoring stated content. The model’s yes/no importance report distinguishes useful concepts among unstated candidates alone, yet falls below chance on the standard pool, where useful concepts are usually unstated. Its workspace readout ranks them above chance there. Self-Distilled Admission (SDA) ranks candidates using a frozen model’s workspace readout and output likelihood, labels the top two, and trains an admission-only LoRA adapter through its importance prompt. Its gradient updates use only labels the frozen model generates itself, with no future question or reward. Across six models, SDA improves answer recall over each model’s own importance report in 54 of 54 model–seed–budget comparisons on the primary statedness-balanced pool. For Qwen3-8B at two stored concepts, SDA adds 10.7 points of answer recall over the model’s own report; published admission criteria (ConsistencyGate, A-MAC, Nemori) score below the untrained report. The adapter runs in a standard serving engine at 242 candidates per second. **The model computes inferences worth keeping, but its importance report fails to follow them; SDA teaches the judgment to keep what it computed.**

1 Introduction

An LLM agent must choose what to remember before it knows the next question. The useful concept is often an inference that the passage never states. A description of a landmark can identify a city without naming it. Remembering that city preserves the link to a later question about the local language. Once the passage leaves the context, the agent’s answer depends on which inferences survived admission to memory.

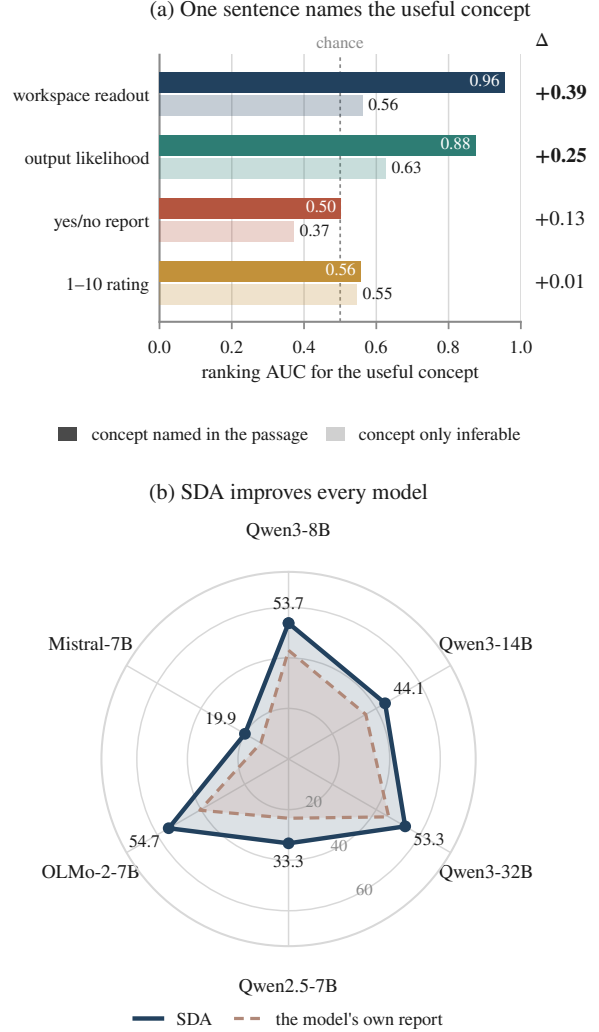


Figure 1: **Naming moves computation far more than importance reports express; Self-Distilled Admission (SDA) turns it into admission decisions.** (a) Six-model mean pooled AUC on the standard Fresh-2 pool, with the useful concept inferable (pale) or named by one added sentence (solid); dashed line: chance. (b) Recall (%) at two stored concepts on the statedness-balanced pool: SDA (solid, three-seed mean) against each model’s own importance report (dashed); mean gain 9.8 points.

Memory systems make this choice through judgments the model expresses. They ask for importance ratings or admit-or-discard decisions (Park et al., 2023), or apply criteria built on context support (Zhang et al., 2026; Zhang and Li, 2026). In each case, the passage provides the evidence for admission. An agent also needs to preserve what it inferred from the passage. This creates a gap between recognizing an inference during reading and selecting it for later use. A memory system can discard the useful concept even when the model has identified it.

We score the same importance report on controlled pools that keep each episode’s useful concept fixed and vary which candidates the passage states. A drop in AUC therefore isolates the effect of provenance: whether the passage states a candidate. The report’s useful-concept AUC falls from 0.880–0.897 among unstated candidates alone to 0.658–0.736 on the statedness-balanced pool and 0.322–0.427 on the standard pool. The statedness-balanced pool states the useful concept in half the episodes. In the standard pool, the useful concept is usually the unstated candidate. The report thus ranks useful concepts below chance when they compete with stated distractors. The 1-10 importance rating and published criteria built on context support show the same preference for stated candidates.

The model’s workspace readout ranks the useful concept above chance on that same standard pool. Its internal computation carries information that its importance report fails to preserve. Naming the useful concept in an otherwise unchanged episode raises AUC for both signals, with a larger increase for the workspace readout (Figure 1). These controlled changes show how admission depends on the way a concept appears in the passage.

Self-Distilled Admission (SDA) transfers the model’s own candidate rankings into its admission decision (Figure 3). A frozen model ranks each episode’s candidates using its workspace readout and output likelihood. It labels the top two candidates for admission, then trains a LoRA adapter (Hu et al., 2021) on those labels through its own importance prompt. Optimization uses only the passage, candidates, and labels generated by the model. It uses no future question, reward, or external teacher. We select the earliest checkpoint by validation admission AUC on a development set with construction-assigned utility labels.

SDA improves recall over the judgments used by existing memory systems. On the primary statedness-balanced pool, it exceeds each model’s own importance report in 54 of 54 comparisons across models, seeds, and memory budgets, and exceeds the 1-10 importance rating in 52 of 54 (Table 1). For Qwen3-8B at two stored concepts, SDA adds 10.7 points of recall over the importance report and 5.2 over the rating. ConsistencyGate, A-MAC, and Nemori fall below the untrained report. SDA exceeds these published admission criteria by 14.1–27.8 points at the same memory budget (Table 2).

The paired-passage battery tests whether the teacher uses the passage to identify the useful concept. Only the passage distinguishes which concept is useful. Across five models, removing the workspace readout from the teacher costs 1.9 recall points at one stored concept (Table 4). Both teacher signals contribute to recall on this battery.

SDA serves as a plug-and-play adapter in a standard engine. It attaches only to admission requests. The served Qwen3-8B adapter reaches 50.1% recall on the standard pool at 242 candidates per second (Table 5).

Our contributions are:

- **Provenance in importance judgments.** We show that importance judgments order candidates by whether the passage states them. Controlled pools separate this preference from usefulness, while the workspace readout tracks useful unstated inferences.
- **Plug-and-play SDA module.** SDA distills the model’s workspace readout and output likelihood into its own importance prompt. The resulting adapter brings these rankings to admission requests through standard serving.
- **Results across models and memory settings.** Evaluation spans six models from three families, three seeds, three memory budgets, and five candidate pools. Comparisons with the importance report, ratings, and published admission criteria establish recall gains. Ablations on the paired-passage battery, where only the passage identifies the useful concept, show that both internal signals in SDA’s teacher contribute to recall.

2 Background and Related Work

Memory systems and evaluation. Generative Agents combines ratings of importance with recency and relevance for retrieval (Park et al., 2023). MemoryBank, MemGPT, Mem0 and A-MEM provide persistence, memory operations and organization (Zhong et al., 2024; Packer et al., 2023; Chhikara et al., 2025; Xu et al., 2025). LoCoMo and LongMemEval evaluate memory over long conversations (Maharana et al., 2024; Wu et al., 2025). SDA targets admission before a future question, which we study through controlled tests of how provenance shapes the importance report.

Admission criteria. ConsistencyGate filters candidates by self-consistent support in the source context (Zhang and Li, 2026), while A-MAC combines LLM assessment with textual support and memory attributes (Zhang et al., 2026). A multi-factor value model optimizes evidence retention (Chen and Cheng, 2026), and Causal Memory Intervention scores answers to a known request under memory interventions (Srivastava, 2026). SDA derives admission targets from the frozen host model’s workspace readout and output likelihood before a future question is available.

Learned memory management. Nemori distills interaction sequences into memory using prediction error (Ma et al., 2026). MemCon learns a policy for retrieval, consolidation and forgetting (Jiang et al., 2026), while AgeMem trains memory tool actions with reinforcement learning (Yu et al., 2026). The survey of Hu et al. (2025) situates these approaches within memory systems. SDA uses the same model’s candidate rankings to supervise admission for a single passage.

Internal signals and representation readouts. EM-LLM, SirLLM and Titans use surprisal, entropy and gradient-based surprise for memory (Fountas et al., 2025; Yao et al., 2024; Behrouz et al., 2025). The logit lens and J-lens decode internal representations (nostalgebraist, 2020; Gurnee et al., 2026), as do tuned lenses, Future Lens and Patchscopes (Belrose et al., 2023; Pal et al., 2023; Ghandeharioun et al., 2024). SDA builds its workspace readout on the logit lens and J-lens, then uses it offline to train a served admission interface.

Memory distillation and self-distillation. Knowledge distillation transfers teacher predictions to a student, including between models of the

same architecture (Hinton et al., 2015; Furlanello et al., 2018). TMEM writes distilled experience into low-rank weights (Ren et al., 2026), while Agent Memory Distillation transfers a larger teacher agent’s trajectories into structured memory for a smaller student (Kim et al., 2026). SDA transfers a frozen model’s candidate rankings into its own importance prompt, keeping teacher and student within the same model.

Internal information and model reports. Internal states can predict correctness beyond expressed confidence (Kadavath et al., 2022; Azaria and Mitchell, 2023; Orgad et al., 2025). Stated reasoning can omit influences on an answer (Turpin et al., 2023), and salience can be poorly reported (Tienes et al., 2025). Introspection training teaches models to predict their own behavior (Binder et al., 2025). SDA trains the importance report to use internal rankings for a prospective decision: which concept to store before the question arrives.

3 The Admission Setting

An agent must select memories before the future question arrives. Our setting makes a useful inference compete with concepts drawn from a passage for a fixed number of memory slots. Each episode contains a passage, an ordered candidate set, utility labels, and a future question with its answer (Figure 2). Only the passage and candidates reach the admission policy.

3.1 Episodes

Each episode centers on a bridge concept: information that links the passage to a future answer. **Decoupled** requires two inference steps: recover the bridge from the passage, then combine it with general knowledge to answer the future question. GPT-4.1 authors a passage about a named person, together with the bridge, question, and answer. The answer is excluded from the passage and candidate set. The question connects to the passage through the person’s name.

The author assigns the positive utility label to the intended bridge. Vivid surface distractors and neutral passage concepts receive negative utility labels. Mechanical filters check candidate strings, bridge and answer exclusion, and question overlap. Deduplication removes repeated concepts. A blind alias audit checks that recovering the bridge still leaves an inference to the answer (App. A).



Figure 2: Memory admission within an episode. The policy receives a passage and candidate concepts before the future question arrives. It stores k concepts. A frozen reader later answers the question from those concepts alone. Here, the useful concept, *dubai*, is the only candidate the passage never states. Figure 5 gives the episode in full. *Drawing guide* (6.3 in $\times$ 2.1 in, Figure 1 palette): five stages left to right, joined by thin arrows. (1) Passage card with two or three lines of Jibril’s passage, “Burj Khalifa” lightly highlighted. (2) Six candidate chips: *elevator*, *desert*, *gift*, *observation and tourists* in grey (stated); *dubai* in light blue (unstated) with a gold star (useful); a small legend. (3) An “Admission policy” box feeding a memory with $k = 2$ slots that hold *dubai* and *elevator*. (4) A dashed vertical line labelled “later”, then a question bubble: “What language was Jibril most likely to hear from local staff?” (5) A frozen reader (grey box with a snowflake) that reads only the two memory slots and answers “arabic”. Small grey notes under the arrows: “policy sees passage and candidates; question hidden” and “reader sees memory only”.

With two memory slots, the city in Figure 2 competes with the elevator, gift shop, and scenery. **Explicit** states the useful content. **Compositional** requires a multi-hop answer and permits passage anchors in the question. **Fresh-1** and **Fresh-2** repeat the Decoupled construction with new passages and bridges absent from the training batteries. Each battery is frozen before measurement.

The author uses the bridge and question to identify the intended memory. The admission policy must predict its value before seeing the question. Comparing admission policies changes the stored concepts while keeping the future question and reader fixed.

3.2 Candidate Pools and Controls

The pools and controls separate the effects of passage mention, candidate competition, context, authorship, and bridge category.

Mention and Position. The **stated-bridge control** appends *The passage also mentions bridge*. The **position control** places that sentence before the passage. The **distractor-named control** substitutes a distractor for the bridge in the appended sentence. Candidates, utility labels, question, and answer stay fixed. These controls isolate the named concept and sentence position.

Competition for Memory. Our primary **statedness-balanced pool** starts from the 363 held-out Fresh-2 episodes in their frozen order. Counting from the first episode, odd positions keep the passage unchanged, giving the inferred-concept condition. Even positions append *The passage also mentions bridge.*, with *bridge* replaced by the useful concept. This gives the stated-concept condition.

Every episode retains its original candidates, including the stated distractors. Odd positions gain four unstated distractors; even positions gain three. These are the first four or three additional candidates, in order, from the pool with unstated distractors described below. Both the naming intervention and the distractor count follow position in the frozen order, with no sampling. The probe question and answer, original utility labels, frozen reader, and blind judge stay fixed. Added candidates receive negative utility labels.

The resulting 363 episodes average 7.93 candidates. Useful concepts account for 13.3% of stated candidates and 12.0% of unstated candidates. A statedness-only rule ranks near chance, with AUC of 0.515 pooled and 0.516 within episodes.

The **naming-balanced pool** names four candidates in every episode, using five sentence tem-

plates at varying positions. Named candidates are useful 12.5% of the time, against a 12.6% base rate.

These pools control different cues: statedness in the statedness-balanced pool and intervention naming in the naming-balanced pool. Their residual regularities point in opposite directions. Intervention naming favors the useful concept in the statedness-balanced pool. Overall statedness favors distractors in the naming-balanced pool.

Two further pools increase competition for the same memory budget. Both start from the standard Fresh-2 pool’s 4.43 candidates per episode. The **pool with unstated distractors** adds four confusable concepts from the bridge’s semantic category. An author model proposes these concepts, which are filtered for absence from the passage and original pool. The **enlarged pool** adds further concepts from other episodes, reaching 12.41 candidates on average. Both additions preserve the original passage, candidates, utility labels, and future question while increasing the alternatives that admission must rank.

Context, Authorship, and Bridge Category.

These controls test whether admission depends on the paired passage, the episode author, or the kind of useful concept. **Shuffled context** replaces the passage with another episode’s passage while retaining its candidates and labels. The **alternate-author benchmark** uses GPT-4o to write episodes with an implied useful concept. The **entity-bridge benchmark** uses GPT-5.6 to place a named-entity bridge among scene-object distractors (App. G.3).

4 Evaluation Setup

Given a passage x and ordered candidates $C = (c_1, \dots, c_m)$, admission stores $K \subseteq C$ with $|K| \leq k$. The future question q and answer a are hidden. When q arrives, the frozen reader sees only q and K ; the original passage has left its window. Every policy receives the same candidates and budget. We measure three successive outcomes: within-episode ranking, retention of the useful concept, and downstream recall.

Within-Episode Ranking. Within-episode area under the receiver operating characteristic curve (AUC) measures how well a policy orders candidates that compete for memory. It is the fraction of useful-distractor pairs ordered correctly, with half credit for ties, averaged across episodes. For useful

concept b_i in episode i , score s and n episodes,

$$A_{\text{episode}} = \frac{1}{n} \sum_{i=1}^n \frac{1}{|C_i| - 1} \times \sum_{c \in C_i \setminus \{b_i\}} \left[\mathbf{1}\{s(b_i) > s(c)\} + \frac{1}{2} \mathbf{1}\{s(b_i) = s(c)\} \right]. \quad (1)$$

Pooled AUC compares candidates across the entire battery and summarizes score discrimination.

For the provenance analysis, we replace utility labels with statedness labels. The resulting *provenance AUC* measures discrimination of stated from unstated candidates. Statedness is determined by a case-insensitive stem match against the passage.

Useful Concept Retention. Bridge retention is the fraction of episodes whose stored set contains the construction-labelled useful concept:

$$B_k = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{b_i \in K_i\}. \quad (2)$$

This measures whether the useful concept survives the memory budget. A policy can rank it above most distractors and still omit it at a small budget. The intended-bridge reference prioritizes the intended useful concept and provides a reference for this construction.

Downstream Recall. Recall is the fraction of future questions that a fixed, frozen reader answers correctly from the stored concepts. A retained useful concept can still lead to an incorrect answer. Correct answering also depends on the reader and the other stored concepts.

The reader answers greedily, with Qwen3 thinking disabled. A blind semantic judge receives only the question, gold answer and response. It accepts a committed correct answer or exact equivalent. A separate string-match grader checks agreement (App. D.1).

Question-only and whole-passage readers establish how much of the answer can be recovered before admission and with the complete original input, respectively.

Comparisons. We compare admission policies under the same inputs and memory budget. Baselines include the untrained yes/no importance report, a 1-10 importance rating, the reversed yes/no importance report, admit what the passage omits,

omission then report, wording variants, and the provenance-adjusted report. We also compare the combined teacher used directly, random selection, output likelihood, the workspace readout, and students trained from either teacher signal alone. All receive the passage with the future question hidden (App. B.4).

Paired AUC differences use 4,000-draw whole-episode bootstraps and 95% intervals. Recall comparisons use exact McNemar tests on paired answers and seed-averaged differences with paired 95% bootstrap intervals (Table 14). Trained results give three-seed means and sample standard deviations. Bold table entries identify SDA.

Evaluation scale. The principal diagnostic comparison covers eight models from four families: Qwen2.5-7B, OLMo-2-7B, Mistral-7B and Qwen3 at 1.7B, 4B, 8B, 14B and 32B (Qwen Team, 2025; Walsh et al., 2025; Jiang et al., 2023; Yang et al., 2025). The extended grid covers 23 checkpoints across those families and GPT-2, including base and post-training stages (App. G.2).

SDA training covers six models with three seeds each: Qwen3-8B, 14B and 32B and the other three 7B families. This yields 18 principal training runs. Each model supplies its own teacher, adapter and frozen reader.

The training and validation splits contain 175 and 45 episodes, respectively. These come from 88 Explicit episodes and 132 implied-answer episodes, including the 57-episode alternate-author benchmark. In an implied-answer episode, the passage cues an unstated useful concept that supplies the answer. Decoupled requires a further inference from that concept.

Development uses Decoupled (68 episodes), Compositional (52), and two separate 68-episode pools that add confusable or unrelated unstated distractors to Decoupled. Fresh-1 and Fresh-2 contain 316 and 363 episodes, respectively. The entity-bridge benchmark contains 42 episodes. Fresh-2 passage and pool controls reuse its 363 episodes.

Evaluation uses memory budgets of $k = 1, 2, 3$. The main comparison uses the statedness-balanced pool, with $k = 2$ reported across all six models. The paired Fresh-2 grading audit covers 13,068 answers across the 18 SDA-report comparisons. Both graders evaluate every answer.

5 Self-Distilled Admission

Self-Distilled Admission (SDA) trains the importance report to retain useful inferences by distilling a frozen model’s candidate rankings (Figure 3; Algorithm 1).

5.1 Teacher Rankings

The frozen model combines its workspace readout and output likelihood to rank each episode’s candidates.

Workspace readout. The workspace readout measures how readily the model’s vocabulary decoder recovers a concept during context processing (nostalgebraist, 2020; Gurnee et al., 2026). Let $h_\ell(x) \in \mathbb{R}^d$ be the residual state at the final context position after transformer layer ℓ , with L layers in total. Final normalization LN and unembedding W_U give vocabulary logits $z_\ell(x) = W_U \text{LN}(h_\ell(x))$. Writing $\text{rank}_\ell(t) = 1 + |\{v : z_{\ell,v} > z_{\ell,t}\}|$, we take the peak reciprocal rank across layers and first-token forms:

$$W(c; x) = \max_{\substack{1 \leq \ell \leq L \\ t \in T(c)}} \frac{1}{\text{rank}_\ell(t)}. \quad (3)$$

Here $T(c)$ contains the distinct first tokens of the candidate’s bare and leading-space forms, each in original and capitalized form. The maximum captures the candidate’s strongest decoding rank across these layers and forms. Computing the readout requires one forward pass over x , using final-position states from all layers.

Combined teacher. Output likelihood scores the complete candidate by its mean log probability per token: $\text{LL}(c; x) = \log p_{\mathcal{M}}(c \mid x)/|c|$, where $|c|$ is the candidate’s token length. To put workspace readout and output likelihood on a common scale, we map each to mid-rank percentiles within the episode. Tied values receive their mean rank (App. B). The teacher used in the main evaluation averages these percentiles:

$$T(c; x) = \frac{1}{2}(P_W(c; x) + P_{\text{LL}}(c; x)). \quad (4)$$

Here P_S denotes the percentile under readout S . The two highest-ranked candidates receive the binary label $\tilde{y} = 1$ for *yes*; all others receive $\tilde{y} = 0$ for *no*. Candidate order breaks score ties. The frozen model generates every teacher label from the passage and candidates.



Figure 3: Self-Distilled Admission. (a) The frozen model combines workspace readout and output likelihood to rank each episode’s candidates and labels the top two yes. (b) A LoRA adapter learns these labels through the model’s own importance prompt. (c) The serving engine applies the adapter only to admission requests. All other requests, including the reader’s answer, use the unmodified model. *Drawing guide (6.3 in $\times$ 2.45 in, Figure 1 palette): three panels. (a) Teacher: passage x and candidates C enter a frozen model (grey box with a snowflake). Beside it, a small heat strip with layers on the vertical axis (1 at the bottom, L at the top) and one column per candidate, shaded by the reciprocal rank of the candidate’s first token at that layer; a navy dot marks each column’s peak, labelled $W(c; x)$. The output head is labelled $LL(c; x)$ in teal. Both become within-episode percentile bars, averaged into T ; a sorted candidate list tags the top two yes (rust) and the rest no (grey). (b) Student: the same frozen model with a small rust LoRA block, answering the importance prompt “Is c one of the most important things to remember...? yes/no”; a dashed line to the labels of (a) reads “next-token cross-entropy”. (c) Deployment: a host agent sends passage and candidates to a serving engine (vLLM); admission requests run with the adapter on and store the top k in memory; question requests run with the adapter off, and the frozen reader answers.*

5.2 Admission Distillation

A LoRA adapter learns the teacher’s labels through the model’s own importance prompt.

Admission score. The importance prompt $u(c, x)$ asks whether c is one of the most important concepts to remember for possible future questions. It requests a single yes or no (App. B.6). Qwen3 uses thinking-disabled prompts. Let Y, N contain the distinct first tokens of the case and leading-space forms of *yes* and *no*, respectively. The admission score compares their total probabilities:

$$s_{\theta}(c; x) = \log \sum_{t \in Y} p_{\theta}(t \mid u(c, x)) - \log \sum_{t \in N} p_{\theta}(t \mid u(c, x)), \quad (5)$$

$$V_{\theta}(c; x) = \sigma(s_{\theta}(c; x)).$$

The untrained model supplies the importance report baseline V . SDA ranks candidates by the adapter-enabled score s_{θ} .

Training objective. We optimize only the LoRA adapter; base parameters remain frozen. Cross-entropy trains next-token prediction of the teacher’s

target word:

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{(x, c) \in \mathcal{D}} \log p_{\theta}(\tau(\tilde{y}(c; x)) \mid u(c, x)). \quad (6)$$

Here τ maps a positive label to *yes* and a negative label to *no*. The target probability is normalized over the vocabulary. Prompt tokens are masked from the loss (App. B.2).

Training and selection. We train a rank-16 adapter for two epochs and select the earliest checkpoint maximizing in-distribution validation admission AUC. Optimization uses teacher labels; validation uses construction-assigned utility labels. We select the combined teacher by mean recall across development batteries and training seeds, scored by the development run’s automatic grader. The configuration is fixed before Fresh-2 (App. C).

5.3 Standard Serving

SDA serves admission decisions through a standard inference engine. At deployment, it makes one yes/no forward pass per candidate and reads the admission prompt’s output probabilities to compute s_{θ} . The host stores the top- k candidates and

Algorithm 1 Self-Distilled Admission.

Require: frozen model $\mathcal{M}$; training episodes $\mathcal{E}$, each a passage x with candidates C

- 1: **for** each episode $(x, C) \in \mathcal{E}$ **do**
 - 2: **for** each candidate $c \in C$ **do**
 - 3: $W(c; x) \leftarrow$ peak reciprocal rank across layers $\triangleright$ Eq. (3)
 - 4: $LL(c; x) \leftarrow \log p_{\mathcal{M}}(c | x) / |c|$
 - 5: $T(c; x) \leftarrow \frac{1}{2} (P_W(c; x) + P_{LL}(c; x))$ for all $c \in C$ $\triangleright$ Eq. (4)
 - 6: $\tilde{y}(c; x) \leftarrow 1$ for the two highest T , else 0
 - 7: train adapter θ on $\mathcal{L}(\theta)$ over all $(x, c, \tilde{y})$ $\triangleright$ Eq. (6)
 - 8: select the earliest checkpoint with the highest validation admission AUC
 - 9: **Deployment:** given a passage x , candidates C and budget k
 - 10: store the k candidates with the highest $s_{\theta}(c; x) \triangleright$ Eq. (5)
-

manages retrieval. Candidate order breaks score ties.

The adapter is attached only to admission requests in a standard serving engine such as vLLM (Kwon et al., 2023). Prefix caching reuses the shared passage. All other requests, including the frozen reader’s answer generation, use the unmodified model.

6 Results

6.1 Provenance and Importance Judgments

Importance judgments track provenance when stated and unstated candidates compete. Among unstated candidates alone, the importance report ranks useful concepts with pooled AUC 0.880–0.897 across six models, compared with 0.592–0.697 for the workspace readout (Table 16). When both provenance classes compete in the statedness-balanced pool, report AUC falls to 0.658–0.736. On the standard pool, it falls below chance to 0.322–0.427, while provenance AUC is 0.582–0.688. The importance report discriminates usefulness within a provenance class and loses that ordering when provenance classes compete.

The workspace readout ranks useful concepts higher than the importance report on the statedness-balanced pool, with pooled AUC 0.719–0.812. Statedness alone provides little useful-concept discrimination in this pool. A rule that reads only whether the passage states a candidate has AUC 0.515 pooled and 0.516 within episodes across all 363 episodes. The within-episode measure tests the competition that admission must resolve: which candidates from the same episode to store.

Changing what the passage names changes both rankings. Naming the bridge raises mean

workspace AUC from 0.563 to 0.956 and report AUC from 0.373 to 0.504 (Table 20). Naming a distractor lowers both on Qwen3-8B (App. F.4). Wording that says the passage may never mention the useful concept follows the same pattern. Qwen3-8B AUC is 0.537 when the useful concept is inferred, 0.417 when named, and 0.621 on Explicit (App. F.3).

Omission identifies the useful concept as the sole unstated candidate in 95% of standard-pool episodes. At $k = 2$, admit what the passage omits and omission then report both reach 53.4% recall; reversed yes/no report reaches 46.6%. Balancing provenance lowers recall for all three policies because omission also favors distractors. All three then rank below the importance report at $k = 2, 3$; omission then report exceeds it at $k = 1$ (Tables 2, 13 and 14).

SDA trails omission then report on the standard pool by 9.5 points at one stored concept ($[-13.7, -5.3]$). At two stored concepts, the difference is -3.4 points ($[-7.0, +0.4]$). On the statedness-balanced pool, SDA exceeds this rule by 14.6 points at two stored concepts (Tables 14 and 8).

6.2 Admission Results

Self-Distilled Admission (SDA) exceeds each model’s own importance report in all 54 model–seed–budget comparisons on the statedness-balanced pool. At $k = 2$, Qwen3-8B recall rises from 43.0% to 53.7/54.8/52.6% across seeds (Table 1). Across budgets, all three seeds reach exact McNemar $p < 0.05$ in 17 of 18 model–budget cells.

The 1-10 importance rating is the strongest untrained admission policy for Qwen3-8B on this pool. We evaluate it for all six models at all three budgets. SDA exceeds it in 52 of 54 model–seed–budget comparisons. The two exceptions are Qwen3-14B at three stored concepts, where two of three seeds reach 44.9% against the rating’s 45.1%.

Qwen3-8B SDA reaches 40.5/53.7/51.8% recall at $k = 1, 2, 3$ on the statedness-balanced pool. It exceeds implied-content wording, the provenance-adjusted importance report, the combined teacher used directly, and direct output likelihood at every budget (Table 2). Section 6.4 separates the contributions of the teacher’s two signals.

At $k = 2$, SDA gains $+5.2$ recall points over the 1-10 importance rating ($[+1.7, +8.8]$). Its gains are $+9.1$ points over implied-content wording, $+14.6$

Table 1: Recall (%) at $k = 2$ across models on the statedness-balanced pool (7.93 candidates per episode). Rating: 1–10 importance rating (Park et al., 2023); p : largest p -value across seeds against each model’s own yes/no importance report.

Model	Yes/no report	1–10 rating	Workspace readout	SDA (3 seeds)	p vs. report	Intended-bridge reference
Qwen3-8B	43.0	48.5	43.0	53.7 / 54.8 / 52.6	$\leq 10^{-5}$	54.3
Qwen3-14B	35.3	40.0	36.6	44.9 / 44.1 / 43.2	$\leq 10^{-4}$	46.6
Qwen3-32B	45.7	43.5	46.0	52.1 / 54.3 / 53.4	$\leq 10^{-3}$	54.8
Qwen2.5-7B	23.4	26.4	24.2	35.3 / 30.6 / 34.2	$\leq 10^{-3}$	31.7
OLMo-2-7B	40.5	47.7	45.2	54.8 / 54.5 / 54.8	$\leq 10^{-7}$	54.8
Mistral-7B	12.7	16.5	16.2	19.3 / 20.4 / 20.1	$\leq 10^{-3}$	16.8

Table 2: Admission policies on the statedness-balanced pool, Qwen3-8B. Recall (%); ConsistencyGate includes sampled and single-pass forms, and prediction error is candidate surprisal under the passage. App. B.5 details published criteria.

Admission policy	$k = 1$	$k = 2$	$k = 3$
<i>Importance judgments of the untrained model</i>			
Yes/no importance report	30.6	43.0	45.5
1–10 importance rating (Park et al., 2023)	35.8	48.5	50.7
Implied-content wording	31.4	44.6	47.7
<i>Admission criteria from published methods</i>			
ConsistencyGate, sampled (Zhang and Li, 2026)	24.0	35.0	40.8
ConsistencyGate, single pass (Zhang and Li, 2026)	24.5	35.0	42.4
A-MAC utility (Zhang et al., 2026)	30.0	39.7	44.9
Prediction error (Ma et al., 2026)	16.5	25.9	30.3
<i>Provenance rules</i>			
Provenance-adjusted report	36.6	39.1	39.9
Reversed yes/no report	23.7	29.8	33.1
Admit what the passage omits	28.6	32.0	36.4
Omission then report	36.9	39.1	39.9
<i>Internal signals used directly</i>			
Output likelihood used directly	34.7	41.9	41.6
Combined teacher used directly	36.9	44.9	46.3
SDA (mean of 3 seeds)	40.5	53.7	51.8
Intended-bridge reference	47.4	54.3	51.5

over the provenance-adjusted importance report, and +8.8 over the direct combined teacher (Table 8). Paired policy differences average seeds and use 95% intervals from 4,000 paired episode-bootstrap draws.

Published admission criteria also fall below the untrained importance report on the primary pool. We implemented ConsistencyGate (Zhang and Li, 2026), A-MAC utility (Zhang et al., 2026), and Nemori’s prediction-error criterion (Ma et al., 2026) from their papers. Each admits top- k candidates under the same budget protocol (Table 2). At two stored concepts, SDA gains +18.7 recall points over each ConsistencyGate form, +14.1 over A-MAC utility, and +27.8 over prediction error. All 12 paired intervals across these methods and budgets exclude zero. The smallest gain

is +6.9 points over A-MAC utility at three stored concepts ([+2.6, +11.1]; Table 8).

ConsistencyGate and A-MAC utility track whether the passage states a candidate. On the statedness-balanced pool, their correlations with statedness are +0.90 for single-pass ConsistencyGate, +0.85 for sampled ConsistencyGate, and +0.75 for A-MAC utility. On the standard pool, ConsistencyGate’s own 0.7 threshold admits 26.7% of useful candidates against 70.8% of all candidates. Its within-episode useful-concept AUC is 0.133 for single-pass scoring and 0.190 for sampled scoring. These criteria exhibit the provenance ordering found in importance reports.

A fitted provenance offset reproduces the development pool’s omission regularity. Adding a single offset for stated candidates, fitted on development episodes, raises development AUC from 0.304 to

0.978. It lowers statedness-balanced AUC from 0.683 to 0.575. The offset ranks every unstated candidate ahead of every stated candidate, reproducing omission then report. Their recalls match at $k = 2, 3$; at $k = 1$, they reach 36.6% and 36.9%, respectively. The offset ranks every unstated candidate ahead of every stated one on every pool, while SDA’s ranking changes with the passage (Figure 4).

SDA improves retention of inferred useful concepts and retains stated concepts. For Qwen3-8B at two stored concepts, the importance report retains the useful concept in 46.4% of stated episodes and 33.0% of inferred episodes. The workspace read-out reaches 92.8% and 35.7%, respectively. Across three seeds, SDA reaches 76.8/81.8/72.4% in stated episodes and 64.8/64.8/64.3% in inferred episodes.

The 173 inferred-condition episodes contain no sentence naming any candidate. SDA reaches 37.4/50.1/46.6% recall at $k = 1, 2, 3$, compared with 24.9/40.5/37.0% for the importance report. The gains are 12.5 ($[+6.7, +18.7]$), 9.6 ($[+3.7, +15.6]$), and 9.6 ($[+4.4, +15.0]$) points. SDA also exceeds the 1-10 importance rating by 5.6 points at two stored concepts ($[+0.8, +10.8]$). These differences average three seeds, with 95% intervals from 4,000 episode-bootstrap draws.

On the standard pool, all 18 runs exceed their own importance report, with a mean gain of 9.6 points and exact McNemar $p \leq 0.026$ (Tables 12 and 14). Semantic and string-match grading both recover these gains (App. D.1).

Useful-concept retention supplies information that the frozen reader uses. SDA improves Qwen3-8B within-episode ranking and retention on the standard pool (Table 15). We replay SDA’s two selections on the 249 episodes where it retains the bridge, then replace the bridge with the next-ranked candidate. Recall falls from 54.6% to 37.8%, a paired decrease of 16.9 points. The replacement loses 59 answers and gains 17 (exact McNemar $p = 1.4 \times 10^{-6}$).

Reader references and candidate proposal. Question-only and whole-passage readers reach 22.9% and 63.9% recall on Qwen3-8B, bracketing SDA’s 50.0% on the standard pool. With a context-only proposer, SDA reaches 35.5/33.3/33.1% across three seeds at $k = 2$ (Table 3; proposal prompt in App. B.6).

Table 3: Question-only and whole-passage reader references, plus context-only candidate proposal, on Fresh-2, Qwen3-8B. Recall (%); admission stores two concepts. Proposal supplies 8.0 candidates per episode, including the useful concept in 1.4% of episodes.

Condition	Recall
<i>Reader references</i>	
Question only	22.9
Whole passage	63.9
<i>Admission from proposed candidates</i>	
Yes/no importance report	33.6
Intended-bridge reference	34.4
SDA (3 seeds)	35.5 / 33.3 / 33.1

6.3 Robustness Across Candidate Pools

SDA’s recall gains extend to the naming-balanced pool, the pool with unstated distractors, and the enlarged pool. On the naming-balanced pool, SDA exceeds each model’s own importance report in 17 of 18 model–budget cells across six models. The remaining cell differs by 0.1 points (Table 6). SDA exceeds the 1-10 importance rating in all twelve model–budget cells at one and two stored concepts. At three stored concepts, SDA and the rating are within 2.6 points for every model. Qwen3-8B also improves within-episode ranking and useful-concept retention across seeds (Table 7). Admission gains persist when naming carries no utility information.

On the pool with unstated distractors, SDA exceeds each model’s importance report in every model–seed–budget comparison (Table 10). Here, the useful inference competes with other unstated concepts. For Qwen3-8B, SDA also exceeds direct output likelihood at every budget on this pool and the enlarged pool (Tables 13 and 11).

At $k = 2$ on the pool with unstated distractors, SDA gains 9.6 recall points over the importance report, 4.1 over the 1-10 importance rating, and 5.8 over omission then report (Table 14). On the enlarged pool, with 12.41 candidates, the corresponding gains are 11.0, 4.7, and 6.6 points. Every paired 95% interval for these comparisons excludes zero (Table 14).

At one stored concept, SDA trails omission then report by 8.4 points on the pool with unstated distractors ($[-12.7, -4.2]$) and by 5.5 points on the enlarged pool ($[-9.6, -1.5]$) (Table 14). At two and three stored concepts, SDA leads every measured baseline on both pools (App. E.1).

6.4 Ablations

Contribution of the workspace readout. Removing either teacher signal lowers recall on the paired-passages battery, where only the passage identifies the useful concept. We remove either the workspace readout or output likelihood from SDA’s teacher and train the same adapter on the resulting labels. Training episodes and the step budget remain fixed.

Each candidate set appears twice, under two passages with their own questions. The useful concept differs within each pair. Both episodes therefore include their partner’s useful concept, and only the passage identifies which concept is useful. We score 181 pairs on Qwen3-8B, Qwen3-14B, Qwen3-32B, Qwen2.5-7B, and OLMo-2-7B, yielding 1,810 observations per budget (Table 4; per-model results in Table 28).

At one stored concept, removing the workspace readout costs +1.9 recall points ([+1.0, +2.8]) and +7.3 retention points ([+6.1, +8.5]). Removing output likelihood costs +1.3 recall points ([+0.3, +2.3]); retention rises by 1.4 points ([−0.2, +2.9]). Both ablations lower mean recall at every budget. Removing the workspace readout also lowers retention at every budget (Table 4). Costs are SDA minus the trained ablation, averaged over three seeds and pooled over models. The 95% intervals use 4,000 bootstrap draws that re-sample pairs, carrying each pair’s rows across all five models together.

The replaced-passages test measures how the student’s selections depend on the passage (Figure 4). We score each episode with its own passage and another episode’s passage, keeping candidates, labels, and the question fixed. On 363 standard-pool episodes with one stored concept, Qwen3-8B SDA retention falls by 20.9 points ([+14.3, +27.4]). The student trained without the workspace readout loses 2.9 points ([−3.5, +9.2]); the student trained without output likelihood loses 26.5 points ([+20.5, +32.6]) (Figure 4). Without the workspace readout, the student’s ordering barely changes when the passage changes.

On the primary statedness-balanced pool, removing output likelihood costs +1.6 and +1.7 recall points at one and two stored concepts. Removing the workspace readout costs +0.7, +0.6, and +0.3 points at one, two, and three stored concepts; all three intervals include zero (Table 27). This analysis combines five models and 1,815 obser-

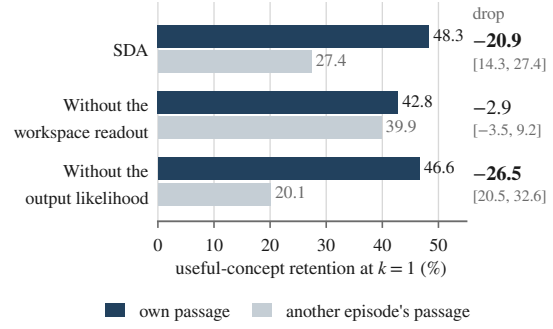


Figure 4: Passage replacement on the standard Fresh-2 pool, Qwen3-8B. Retention is the percentage of episodes whose single stored concept is useful: own passage (solid), another episode’s passage (pale). Drops are own minus replaced retention; Table 30 includes untrained readouts.

ations per budget, with each episode moving together across models in the bootstrap. Table 26 reports per-model recall.

Training the importance interface beyond scoring. Training the importance interface improves recall beyond using the teacher directly. On the standard Fresh-2 pool at $k = 2$, SDA reaches 50.0% recall, compared with 45.7% for the direct combined teacher and 40.2–40.5% for the validation-selected log-odds objectives (Tables 32 and 33; App. H.2).

A matched wording comparison separates training gains from prompt changes. The *implied-content wording* is the strongest measured Qwen3-8B prompt baseline that permits inferences while leaving the useful concept’s provenance unspecified. It raises standard-pool report AUC from 0.321 to 0.462 and improves recall at every budget (Table 17). Holding this wording fixed for training and evaluation, SDA reaches 37.1/50.7/51.0% recall at $k = 1, 2, 3$. It exceeds the prompt baseline at every budget (exact McNemar $p \leq 6.9 \times 10^{-5}$; Table 17). SDA trained with passage-only wording reaches 37.6/50.0/52.2% (Tables 13 and 14).

6.5 Deployment

Standard serving preserves SDA recall at 242 candidates per second on one A100-80GB (Table 5). The served Qwen3-8B adapter reaches 50.1% recall on the standard pool, within 0.3 percentage points of the reference selections. Paired disagreements are two versus three episodes (exact McNemar $p = 1.0$). Top-2 memories agree on 91.7% of

Table 4: Paired-passage battery results for Qwen3-8B, Qwen3-14B, Qwen3-32B, Qwen2.5-7B and OLMo-2-7B. Entries: SDA minus students trained with one teacher signal removed, pooled across models, in percentage points; retention indicates admission of the useful concept.

Budget	Without the workspace readout		Without the output likelihood	
	Retention	Recall	Retention	Recall
$k = 1$	+7.3 [+6.1, +8.5]	+1.9 [+1.0, +2.8]	-1.4 [-2.9, +0.2]	+1.3 [+0.3, +2.3]
$k = 2$	+4.6 [+3.6, +5.7]	+0.6 [-0.2, +1.4]	+0.1 [-1.3, +1.4]	+1.6 [+0.7, +2.5]
$k = 3$	+3.0 [+2.0, +3.9]	+1.1 [+0.4, +1.8]	+1.7 [+0.6, +2.7]	+1.4 [+0.5, +2.3]

episodes, and admission AUC differs by at most 0.001.

The throughput measurement uses batched serving with prefix caching over 1,607 candidates after warm-up. SDA takes 4.1 ms per candidate; the importance report reaches 290 candidates/s at 3.4 ms per candidate (Table 5). The adapter serves admission requests. The frozen model answers from the admitted memory.

7 Implications

Memory admission is a trainable component of an agent. Self-Distilled Admission (SDA) trains the importance interface using rankings from the model’s own computation. A host can attach the adapter only to admission requests and keep the reader frozen. Qwen3-8B serves the standard pool through a standard serving engine (§6.5). SDA improves development admission AUC at every tested Qwen3 training scale and seed. The workspace readout AUC changes by at most 0.0046 (App. G.4).

Admission evaluation should separate usefulness from provenance. Compare policies on the same episodes with a fixed reader and budget. Balance the provenance of the useful concept and include unstated distractors. Report within-episode ranking, retention split by provenance, and frozen-reader recall. Together, these measurements connect candidate scores to the information stored and the answers the reader produces.

8 Conclusion

The importance report orders memory candidates by provenance, favoring stated content even when the useful concept is unstated. Self-Distilled Admission (SDA) uses a frozen model’s workspace readout and output likelihood to train a LoRA adapter through the model’s own importance prompt. Training requires no future questions, rewards, or external teacher. SDA improves re-

call over each model’s importance report on the statedness-balanced pool. The adapter attaches only to admission requests and preserves recall under standard serving. This enables agents to decide what to retain before a question arrives and to store inferred links that later answers need.

9 Limitations

The evaluation covers admission of host-supplied candidates in authored episodes with one later question. End-to-end long-conversation memory also requires proposal, revision, retrieval, and deletion. Construction labels identify an intended bridge rather than the optimal memory for every reader. Stem matching approximates provenance, and semantic grading uses a model judge. Serving and full-candidate readout measurements cover Qwen3-8B on the standard pool. Serving uses batched scoring on one A100-80GB. The objective ablation measures ranking and recall rather than probability calibration, with three seeds at learning rate 10^{-5} and one at each smaller learning rate. The 54 model–seed–budget comparisons share episodes. They measure consistency across settings rather than 54 independent replications.

Table 5: Admission serving on Qwen3-8B, standard pool, one A100-80GB with prefix caching. Amortized time is per candidate; top-2 selections match the reference on 91.7% of episodes, with admission AUC differing by at most 0.001.

Admission scoring	Time (ms/candidate)	Candidates/s	Hidden states
Yes/no importance report	3.4	290	not needed
SDA adapter	4.1	242	not needed

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A Benchmark Construction and Leakage Audit

A.1 Passages, concepts and future questions

The primary author model is GPT-4.1. Each episode contains a passage, candidate concepts with construction-assigned utility labels, and a future question and answer. Explicit places useful content in the passage. **Implied-answer** training episodes cue an unstated useful concept that itself supplies the answer, including episodes from the alternate-author benchmark. The extended tables use this name for that training construction. Decoupled requires an unstated bridge to support a further inference: the answer is absent from the passage and candidate set, and the question shares no passage content word except a protagonist’s name. Compositional requires a multi-hop inference and permits question anchors from the passage. Decoupled and Compositional are development batteries used for configuration selection and excluded from gradient training.

Mechanical validation checks candidate strings, bridge absence, answer exclusion and the Decoupled question’s lexical overlap. Concept deduplication removes repeated concepts across retained episodes. The leakage audit additionally checks full lexemes, inflected answers and question anchors. A blind GPT-6-Astra alias audit receives the future question, gold answer and bridge. It rejects identity, morphological or derivational variants, synonyms, translations, abbreviations and containment that trivially yields the answer. For example, *caterer/catering* is an alias, whereas *Madrid/Spanish* requires a further inference. The audit records the verdict and exclusion reason for each episode.

A.2 Episode-authoring prompt

GPT-4.1 receives the system instruction, “You are a careful dataset designer for a cognitive-science experiment on LLM memory.” The following prose transcription preserves the generation constraints (Prompt 1); the output-format instruction requests one structured record per episode. Generation uses temperature 1.0 and batches of up to five episodes. Mechanical filters and the alias audit apply after generation.

A.3 Frozen batteries and replication benchmarks

Fresh-1 and Fresh-2 use the Decoupled construction prompt and mechanical filters, followed by deduplication and disjointness filtering. A context match after lowercasing and trimming against a prior battery causes exclusion. A bridge matching a construction-labeled training concept also causes exclusion. Fresh-2 additionally excludes contexts appearing in Fresh-1. Shared answer categories, such as languages and currencies, are permitted. The retained batteries contain 316 and 363 episodes, respectively. The alternate-author benchmark repeats the inferred-concept construction with a different author model. The entity-bridge benchmark places a named-entity bridge among scene-object distractors.

A.4 Reconstructing the reported controls

All pool interventions traverse the 363 held-out Fresh-2 episodes in their frozen order and preserve the question, answer and original utility labels. Added candidates receive negative utility labels. Within an episode, original candidates precede additions, whose recorded order is retained. The frozen control archive records the passages, ordered additions, named candidates and context permutation; replay uses these recorded draws. The standard pool contains 1,607 candidates, 4.43 per episode. Statedness uses the case-insensitive stem match specified in §4.

Worked source episodes. Figure 5 gives the first two Fresh-2 episodes in full; the second is the episode drawn in Figure 2.

Stated-bridge, position and distractor-named controls. The exact sentence form is “The passage also mentions *concept*.” The stated-bridge control appends this sentence with the useful concept; the position control prepends it. The distractor-named control appends it with the first non-useful candidate in the original order. Thus Yara’s stated-bridge control adds “The passage also mentions insurance.” after the passage, the position control puts that sentence before “Yara compared,” and the distractor-named control instead appends “The passage also mentions insurer.” These three controls keep all five candidates and their order unchanged.

Pool with unstated distractors. An author model proposes four single-word concepts from

Example 1: First Fresh-2 episode Passage. Yara compared renewal quotes from different insurers online and reviewed her accident history. She called the hotline to ask clarifying questions about premium increases. After considering the coverage limits and deductibles, Yara finally entered her payment details to confirm the policy. Candidates, in supplied order. insurer, deductible, history, policy, insurance. Useful concept. insurance. Surface distractors. deductible, history. Neutral concepts. insurer, policy. Future question. Which professional would be most relevant for Yara to contact regarding her recent purchase? Answer. agent.	Example 2: Second Fresh-2 episode Passage. Jibril had always wanted to gaze up at the world’s highest outdoor observation deck. Early one morning, he took a swift elevator to level 148 of the Burj Khalifa, joining a crowd of eager tourists. Below him, the city sparkled with the first rays of the desert sun. After snapping several panoramic photos, he browsed the sleek gift shop for souvenirs. Candidates, in supplied order. elevator, desert, gift, observation, dubai, tourists. Useful concept. dubai. Surface distractors. desert, tourists. Neutral concepts. elevator, gift, observation. Future question. What language was Jibril most likely to hear from local staff during his visit? Answer. arabic.
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Figure 5: First two Fresh-2 episodes in full. Admission sees only the passage and candidate strings; utility labels, questions and answers are included for explanation.

the useful concept’s semantic category, asking for alternatives a careless reader might confuse with it but that the passage does not imply. The instruction requires lowercase words absent from the passage and a comma-separated response. We retain the response order, discard a proposal that matches an original candidate or occurs as a case-insensitive substring of the passage, and append the accepted concepts. The frozen pool contains 1,450 additions: four in 361 episodes and three in episodes 24 and 263 after filtering, giving 3,057 candidates in total (8.42 per episode). For Yara, the ordered additions are *warranty, mortgage, pension, annuity*; her passage and five original candidates stay fixed. The development confusable-distractor control applies the same procedure to Decoupled. The unrelated-distractor control instead takes bridges from subsequent episodes, as described for the enlarged pool below.

Statedness-balanced pool. Odd positions, counting from one, keep the original passage; even positions append “The passage also mentions *concept*.” with the useful concept substituted. Every episode keeps its original candidates and takes the first four accepted additions at odd positions or the first three at even positions from the pool with unstated distractors, preserving their order. This yields 2,877 candidates, 7.93 per episode. Yara occupies position one, so her passage is unchanged and her pool is the five original candidates followed by *warranty, mortgage, pension, annuity*. Jibril occupies position two: append “The passage also mentions dubai.” to Example 2’s passage and append *doha,*

riyadh, muscat to its six candidates. The original question and answer remain fixed in each case.

Naming-balanced pool. Use the same ordered candidate lists as the statedness-balanced pool, but start from the original passages. Name four candidates per episode. At odd positions exclude the useful concept and sample four other candidates uniformly without replacement; at even positions include the useful concept and sample three others. Use one Python standard pseudorandom generator seeded with zero, process episodes in frozen order, prepend the useful concept to the sampled list when included, and shuffle each selected list before constructing sentences. Sampling and shuffling consume the same continuing random stream. The five sentence forms, in rotation, are “The passage also mentions *concept*.”, “*concept* is relevant here.”, “Note that *concept* comes up as well.”, “There is also *concept* to consider.”, and “*concept* features in this as well.”

Number episodes and insertions from zero for the placement rule. The first selected candidate uses the sentence form at the episode’s position in the five-form cycle; each subsequent candidate advances one form. Strip trailing full stops from the original passage, split at full-stop–space boundaries, trim each segment and restore one full stop to each. Before each insertion, count the current sentence list, including earlier additions. The insertion boundary is the remainder of the episode index plus three times the insertion index upon division by that count plus one. Boundary zero precedes the passage; the final boundary follows it. Inser-

tions thus advance through an evolving sentence list rather than all occurring at its end.

For Yara, the shuffled selection is *pension, annuity, insurer, policy*. The additions are “The passage also mentions pension.”, “annuity is relevant here.”, “Note that insurer comes up as well.”, and “There is also policy to consider.” Their successive insertion boundaries are zero, three, zero and two. The final passage begins with the insurer, pension and policy sentences, then Yara’s first two original sentences, then the annuity sentence, then her final original sentence. There are 1,452 naming interventions, 181 of which name the useful concept: a named candidate is useful 12.5% of the time, against the 12.6% base rate among 2,877 candidates. Questions, answers and candidate order stay fixed.

Enlarged pool. Combine the standard candidate list, the accepted same-topic additions, and four bridges from other episodes. To choose the latter, start with the next episode in frozen order, advance cyclically, and accept a bridge only if it is absent from the passage, absent from the original candidate list and distinct from earlier accepted bridges. Stop after four accepted bridges. Concatenate the three lists in this order and retain the first occurrence of each concept. The resulting 4,506 candidates average 12.41 per episode. Yara’s pool is *insurer, deductible, history, policy, insurance, warranty, mortgage, pension, annuity, dubai, church, kanji, doctor*. Her passage, question and answer remain unchanged. The unrelated-distractor control uses just the standard candidates and the four cross-episode additions; for Yara these are *dubai, church, kanji, doctor*.

Shuffled context and bridge replacement. Shuffled context uses the archived episode permutation, replacing each passage while preserving its own candidates, utility labels, question and answer. The first episode receives the passage at position 105, about Lars skating on slick tracks with thin-bladed boots; its candidates remain Yara’s five concepts and its answer remains *agent*. The bridge-replacement control acts on SDA’s recorded top-two selections only when they contain the useful concept: retain the other selection and replace the useful concept with the next-ranked candidate, then replay the same frozen reader and question. For a selection of *dubai, elevator* with *gift* next, the replayed memory is *gift, elevator*; Jibril’s question and *arabic* answer stay fixed. This example

illustrates the replacement rule; the measured 249 replays use the recorded selections (§E.2).

Candidate proposal and reader references. The context-only proposer receives the original passage and the instruction in Appendix B.6; the future question and bridge are hidden. Its parsed output replaces the supplied candidate list in proposal order. For Yara the resulting candidates are *online, accident, hotline, premium, coverage, deductibles, payment, policy*, so *insurance* is unavailable to admission. At two stored concepts, the same frozen reader answers from each policy’s selected pair. The question-only reference gives that reader Yara’s question with no memory; the whole-passage reference gives the same question and her original three-sentence passage. Neither reference performs admission. The wording controls change only the admission instruction, whose five exact forms appear in Appendix F.3. Authorship and entity-bridge controls repeat the episode construction with the authors and concept categories specified in Appendix G.3.

SDA improves ranking and retention under the naming-balanced construction (Table 7). Across Qwen3-8B seeds, within-episode AUC is 0.715–0.725, versus 0.566 for the report and 0.636 for the workspace readout. Useful-concept retention at $k = 2$ is 53.2–54.0%, versus 24.0% and 40.8%, respectively.

B SDA Training and Scoring Details

B.1 Teacher construction and data split

Within each episode, let $r_S(c)$ be a candidate’s ascending mid-rank under readout S , with ties assigned the average of their occupied ranks. For $m > 1$ candidates, the percentile is

$$P_S(c; x) = \frac{r_S(c) - 1}{m - 1}, \quad S \in \{W, LL\}. \quad (7)$$

For a singleton pool the percentile is one half. Averaging the percentiles yields the teacher in Eq. (4); descending teacher score and then candidate order determine the top-2 labels. Both frozen readouts score the same episode and candidate before these labels are constructed.

A source group is one author-and-construction battery: Explicit (88 episodes), implied-answer from the primary author (75), or implied-answer from the alternate author (57). The split is stratified by these groups and assigns whole episodes, including all their candidates, to training or validation.

Table 6: Recall (%) on the naming-balanced pool, six models, at one, two and three stored concepts. Report: each model’s own yes/no importance report; Rating: 1–10 importance rating (Park et al., 2023).

Model	$k = 1$			$k = 2$			$k = 3$		
	Report	Rating	SDA	Report	Rating	SDA	Report	Rating	SDA
Qwen3-8B	27.6	35.3	36.4	42.7	45.2	46.2	46.3	47.1	47.0
Qwen3-14B	21.8	30.3	33.7	33.3	36.4	40.8	41.3	43.2	41.2
Qwen3-32B	31.7	30.0	41.1	43.2	43.0	48.9	50.7	51.8	53.4
Qwen2.5-7B	9.4	11.6	13.3	21.2	25.6	28.3	28.6	33.3	33.0
OLMo-2-7B	24.8	30.6	37.0	39.4	43.0	46.7	44.6	52.1	49.5
Mistral-7B	4.7	5.8	9.9	9.6	13.2	16.0	19.8	21.5	23.7

Table 7: Candidate ranking and useful-concept retention on the naming-balanced pool, Qwen3-8B, 2,877 candidates. Entries are AUC and retention (%).

Admission policy	Pooled AUC	Episode AUC	B_1	B_2	B_3
Yes/no importance report	0.570	0.566	9.4	24.0	42.7
Workspace readout	0.635	0.636	22.0	40.8	54.8
SDA, seed 0	0.726	0.725	36.1	54.0	67.5
SDA, seed 1	0.714	0.715	32.2	54.0	65.6
SDA, seed 2	0.728	0.725	35.5	53.2	68.0

Within each group, episodes are sorted by a cryptographic hash of the fixed split seed and the source-qualified episode identity; the lowest 20%, rounded up, enter validation. The split seed is zero and stays fixed across optimization seeds. This gives 70/18, 60/15 and 45/12 training/validation episodes in the three groups, respectively: 175 training and 45 validation episodes in total. The teacher generates every optimization target from these passages and candidates. The selection protocol is specified in §5.2.

Setting	Configuration
Adapter	Low-rank adaptation, rank 16, scaling 32; attention and feed-forward projections
Optimizer	AdamW; constant learning rate 1×10^{-5}
Batching	Batch size 1; gradient accumulation over 4 batches
Training duration	Two epochs; 414 optimizer steps
Sequence length	At most 1024 tokens
Data	175 training and 45 validation episodes; source-stratified, disjoint episodes
Targets	Two highest teacher ranks positive; all others negative

B.2 Complete training configuration

LoRA adapts the attention query, key, value and output projections and the feed-forward gate, up and down projections. Base parameters remain frozen. Each model supplies its own teacher rankings and receives its own admission adapter. The two highest-ranked candidates in each training episode receive positive admission targets; all remaining candidates receive negative targets. Only the target word contributes to the vocabulary-normalized cross-entropy in Eq. (6). Prompt tokens are masked. Qwen3 uses its chat template with thinking disabled.

B.3 Measurement conventions

The yes/no report in every main table is scored in log space over the yes and no token sets. The wording comparison uses its own common scorer for all five wordings, so its rows are comparable to one another under that aggregation: Mistral-7B’s passage-only wording gives AUC 0.411 in the main tables and 0.440 in the wording comparison because the two scorers aggregate differently; scoring the same wording one prompt at a time also gives 0.440, identical to the batched result. On Qwen3-8B’s standard pool, first-token workspace AUC is 0.543 at the end of the passage and 0.530 when the candidate is appended and the state is read immediately before its first token; the latter is the matched baseline for full-candidate AUC 0.574. Frozen-report recall is 40.8% in the main tables and 40.5% in the objective ablation, which evaluates the

untrained checkpoint through the adapter-loading path instead of the frozen path.

B.4 Admission baselines

SDA is compared with the following baselines. The **yes/no importance report** uses the untrained model’s score V in Eq. (5), with Qwen3 thinking disabled. The **1–10 importance rating** is the expected rating under normalized probabilities of the complete answer strings. The **reversed yes/no report** admits the lowest-scoring candidates. **Admit what the passage omits** prioritizes candidates absent from the passage under the statedness test and breaks ties at random. **Omission then report** prioritizes unstated candidates and uses the yes/no importance report to order candidates within each provenance. The **provenance-adjusted report** adds a single coefficient times the candidate’s statedness indicator to the untrained report, with the coefficient fitted on development episodes and held fixed for evaluation. The **combined teacher used directly** ranks by Eq. (4) without training an adapter. The **implied-content wording** permits implied content; the **provenance-cued wording** explicitly says the passage may never mention the useful concept (App. F.3). **Random selection** samples candidates uniformly; **output likelihood** and the **workspace readout** rank candidates directly by their respective scores. Each baseline sees the passage and candidates with the future question hidden. **intended-bridge reference** prioritizes the intended bridge; question-only and whole-passage readers provide recall references. Bold entries identify SDA throughout the comparison tables.

B.5 Published admission criteria as implemented

ConsistencyGate. ConsistencyGate (Zhang and Li, 2026) is a write-time gate that scores how strongly the source context supports a candidate fact. We measure two forms. The sampled form follows their Equations 2 and 3, using their Appendix C prompt verbatim with five samples at temperature one and at most eight new tokens per sample. Its score is the mean of the parsed decimals. The single-pass form follows their Equation 4 variant: the same question is posed as a yes/no question, and the score is the probability of yes from a softmax over the yes and no token log-probabilities. Their threshold of 0.7 is used only for the gate-rate diagnostic. Threshold-based admission is not used

for ranking; ranking uses the score itself, giving the method the same budget as every other policy.

A-MAC. A-MAC (Zhang et al., 2026) is a multi-factor value model. Utility is obtained from one temperature-zero call asking whether the candidate is actionable, supports likely follow-up questions, or captures a persistent constraint or preference. Confidence is the maximum ROUGE-L between the candidate and a supporting passage sentence. The type prior is their rule-based entity cue. Their novelty term compares against memory already held, and their recency term decays with elapsed time. Within an episode, memory is empty at the first admission and all candidates share one times-tamp, so neither term orders candidates. These two terms are therefore omitted from the ranking score. The three remaining factors are combined as a weighted sum, with the utility weight fixed at one. Confidence and type-prior weights are searched on a grid from 0.0 to 2.0 in steps of 0.1, selected by within-episode AUC inside five-fold cross-validation over the 363 episodes, and evaluated out of fold. Every fold selects zero weight for both confidence and type prior, so the fitted combination coincides with utility alone. Out-of-fold within-episode AUC is 0.6795 for both.

Nemori. Nemori’s prediction-error retention criterion (Ma et al., 2026) keeps what the model fails to predict. Applied to a candidate concept and its passage, the score is the candidate’s surprisal under the passage, the negative of the output likelihood. The criterion is used to rank candidates for admission; retention of already-held memories is not exercised in this candidate-admission setting.

All three score the same unit as every other policy, a candidate concept and its episode passage with the future question hidden, and admit the top- k under the same budget protocol. Prompts 4 to 6 give the three scoring instructions.

B.6 Admission, reader, rating and proposal scoring

Appendix I reproduces every natural-language instruction. Each is a user message rendered with the model’s chat template and assistant generation prefix; Qwen3 thinking is disabled. SDA and the report baseline share the admission prompt (Prompt 2).

The affirmative token set contains the distinct first tokens of yes in lowercase, initial-capital and uppercase forms, with and without a leading space.

The negative set is constructed analogously from *no*. Equation (5) gives the common report score; SDA applies the same scoring rule with its adapter enabled.

The reader (Prompt 8) uses the frozen model with its admission adapter disabled. The question-only reference asks “Answer this question concisely.” followed by the question and answer prefix. The whole-passage reference supplies the passage, then “Answer concisely.” and the same question and answer prefix. Greedy decoding is used throughout.

Let a_r be the complete answer string for rating r , and let $u_R(c, x)$ be the rating prompt (Prompt 3). Each answer receives its full sequence probability, multiplying the conditional probabilities of its tokens. The score is

$$R(c; x) = \sum_{r=1}^{10} r \pi_r(c; x),$$

$$\pi_r(c; x) = \frac{p_{\mathcal{M}}(a_r \mid u_R(c, x))}{\sum_{j=1}^{10} p_{\mathcal{M}}(a_j \mid u_R(c, x))}. \quad (8)$$

A rating represented by multiple tokens is scored as a complete answer.

The context-only proposer (Prompt 7) uses Qwen3-8B with greedy decoding. Distinct parsed concepts form the candidate pool. The bridge, future question and answer are withheld from proposal and carried forward for evaluation. Bridge coverage accepts case differences and a trailing plural ending.

B.7 Ablation protocol

The teacher and optimization ablations in §6.4 use the same training episodes, adapter shape and matched step budget. The recall-reward condition samples admitted sets from the admission scores and asks the frozen reader to answer from the selected memory. Answer correctness supplies the reward. Within each rollout group, reward differences are normalized into advantages; updates use a clipped group-relative policy objective and a penalty for departure from the frozen policy (Shao et al., 2024). The future question enters the reward computation. The refinement condition applies these updates after SDA training.

C Documented Prospective Freeze

The dated experiment record documents prospective freezes of Fresh-1 and Fresh-2 on 15 September 2026. It records each battery’s episode count

and cryptographic digest before evaluated models read it: 316 episodes for Fresh-1 and 363 for Fresh-2. The accompanying reproducibility record retains the dated entries and verified battery digests. Fresh-1 supplied the initial frozen-battery evaluation. Fresh-2 was frozen before any evaluated model read it and after configuration selection. Its primary endpoint compares SDA with the yes/no report; a primary training seed was designated and the remaining seeds reported alongside it. Its secondary endpoint compares SDA with the rating baseline. Fresh confirmation uses the blind semantic judge in App. D.1.

D Recall Grading and Serving Protocol

D.1 Blind semantic judge

The judge, GPT-6-Astra, receives the question, gold answer and reader response with no admission-policy identity. Prompt 9 quotes the substantive grading instruction verbatim; its output-format instruction requests a label for every supplied answer. Only correct responses contribute to recall; hedged and incorrect responses count as errors. Judgments are reused for identical question, gold answer and response triples. The harness additionally grades each answer by string matching after normalization. Agreement is 85.1% across 13,068 answers; every training run exceeds its own report under both graders, with maximum exact McNemar $p = 0.026$.

D.2 Amortized scoring time and throughput

The timing evaluation loads Qwen3-8B on one A100-80GB and constructs an admission prompt for each of the 1,607 standard-pool Fresh-2 candidates. vLLM enables prefix caching and LoRA serving, warms the engine on an initial subset, and times batched generation over the full candidate list. Each prompt generates one token greedily; SDA attaches the adapter to the admission request, while the report baseline uses the same engine without it. Model loading is outside the timed region. Milliseconds per candidate are wall time divided by candidate count; throughput is candidate count divided by wall time. The served and reference selections are passed to the same frozen reader and semantic judge. The serving engine recovers yes/no masses from returned token log probabilities.

D.3 Capability with the adapter enabled on every token

Enabling the adapter on every token changes the four measured capability scores by at most 0.3 percentage points under matched prompts. Base and adapter scores are 71.6% and 71.5% on MMLU (Hendrycks et al., 2021), 80.4% and 80.7% on ARC-Easy, 56.2% and 56.4% on ARC-Challenge (Clark et al., 2018), and 88.4% and 88.6% on the GSM8K evaluation subset (Cobbe et al., 2021). The deployed interface attaches the adapter to admission requests.

E Recall across Evaluation Batteries and Pools

SDA exceeds the report in all 54 comparisons across six models, three seeds and three budgets. All seeds reach exact McNemar $p < 0.05$ in 17 of 18 model-budget cells; Qwen3-32B at $k = 3$ has maximum $p = 0.073$. Across the six models, pooled report AUC is 0.658–0.736 and workspace AUC is 0.719–0.812.

Qwen3-8B SDA improves mean recall over the report on each development and frozen battery in Table 9. The gains also occur with unstated distractors and in the enlarged pool (Tables 10 and 11). Across all six models, mean SDA recall exceeds the report and rating at every standard-pool budget (Tables 12 and 13).

E.1 Candidate-pool comparisons

The standard pool rewards omission. The useful concept is the only candidate the passage never states in 95% of standard-pool episodes. The omission baseline therefore reaches 46.6%, 53.4% and 49.6% recall at $k = 1, 2, 3$, respectively; omission then report reaches 47.1%, 53.4% and 52.6%, and the reversed report reaches 35.5%, 46.6% and 49.0%. SDA’s corresponding means are 37.6%, 50.0% and 52.2%. The omission cue identifies the useful concept directly in these episodes.

Unstated distractors separate useful admission from omission. The pool with unstated distractors adds four unstated distractors per episode. Under blind semantic grading, SDA exceeds each model’s own yes/no report in 54 of 54 comparisons across six models, three seeds and budgets $k = 1, 2, 3$. All three seeds reach exact McNemar $p < 0.05$ in 17 of 18 model-budget cells; the exception is Mistral-7B at $k = 3$. At $k = 2$, the largest p across models and seeds is 0.019

(Table 10). On Qwen3-8B, direct output likelihood reaches 30.6/39.1/40.5% recall at $k = 1, 2, 3$, against SDA’s 37.6/50.7/51.6% means. SDA leads the measured baselines at $k = 2, 3$; omission then report leads at $k = 1$ (Tables 13 and 14). At $k = 2$, the omission baseline falls from 53.4% to 35.3% and the reversed report from 46.6% to 28.9%, both below the report’s 41.0%; SDA reaches 51.0%, 51.2% and 49.9%. Every SDA seed exceeds the omission baseline at every budget, with exact McNemar $p \leq 0.097$ at $k = 1$ and $p \leq 6.5 \times 10^{-7}$ at $k = 2, 3$; every seed exceeds the reversed report with $p \leq 4.3 \times 10^{-7}$ across budgets. SDA recalls are 37.7% / 36.4% / 38.6% at $k = 1$ and 51.8% / 51.5% / 51.5% at $k = 3$; the rating baseline reaches 35.8% and 48.8%.

Naming the bridge and changing its category test reversal. On the stated-bridge control at $k = 2$, SDA reaches 55.4%, 55.6% and 52.9% recall, against 44.9% for the report and 39.9% for the reversed report. On the entity-bridge benchmark, the reversed report reaches 0.281 admission AUC, while SDA reaches 0.987, 0.980 and 0.987, with within-episode AUC 1.000. The direction required by reversal is a property of candidate construction, and is opposite on this benchmark (§6.1).

The enlarged pool preserves gains at two and three concepts. With 12.41 candidates per episode, SDA reaches 38.0% / 38.0% / 39.7% at $k = 1$, 50.7% / 52.6% / 51.2% at $k = 2$, and 52.9% / 52.1% / 52.1% at $k = 3$ (Table 11). The rating baseline reaches 36.4%, 46.8% and 48.2%, respectively; every SDA seed leads it at every budget. The paired mean differences are +2.2 points ($[-1.3, +5.6]$), +4.7 ($[+0.9, +8.4]$), and +4.1 ($[+0.9, +7.6]$), respectively (Table 14). Omission then report has the highest recall at $k = 1$; SDA leads the measured baselines at $k = 2, 3$. Direct output likelihood reaches 28.1/32.0/33.9%, against SDA’s 38.6/51.5/52.4% means. Random selection reaches 21.5%, 27.0% and 29.8%, the report 29.5%, 40.5% and 41.9%, and the workspace readout 24.0%, 34.7% and 38.6%.

E.2 Within-episode ranking and bridge retention

SDA improves both within-episode ranking and retention of the intended bridge at $k = 2$ (Table 15). Paired whole-episode bootstrap intervals compare each score with the report baseline. Conditional recall distinguishes retaining the bridge from an-

Table 8: Paired SDA minus baseline recall differences (percentage points) for every row of Table 2, statedness-balanced pool, Qwen3-8B. Published methods: 1–10 importance rating (Park et al., 2023); ConsistencyGate (Zhang and Li, 2026); A-MAC (Zhang et al., 2026); prediction error (Ma et al., 2026).

Baseline	$k = 1$	$k = 2$	$k = 3$
<i>Importance judgments of the untrained model</i>			
Yes/no importance report	+9.9 [+6.1, +13.8]	+10.7 [+7.0, +14.6]	+6.3 [+2.9, +9.7]
1–10 importance rating	+4.7 [+1.0, +8.3]	+5.2 [+1.7, +8.8]	+1.1 [−2.4, +4.6]
Implied-content wording	+9.1 [+5.4, +12.9]	+9.1 [+5.7, +12.7]	+4.1 [+1.0, +7.3]
<i>Admission criteria from published methods</i>			
ConsistencyGate, sampled	+16.5 [+11.6, +21.5]	+18.7 [+14.1, +23.6]	+11.0 [+6.3, +15.5]
ConsistencyGate, single pass	+16.0 [+10.8, +21.1]	+18.7 [+14.0, +23.5]	+9.4 [+4.8, +14.1]
A-MAC utility	+10.5 [+5.6, +15.3]	+14.1 [+9.3, +19.0]	+6.9 [+2.6, +11.1]
Prediction error	+24.0 [+18.9, +29.1]	+27.8 [+22.5, +33.2]	+21.5 [+16.4, +26.6]
<i>Provenance rules</i>			
Provenance-adjusted report	+3.9 [−0.9, +8.5]	+14.6 [+9.5, +19.8]	+11.8 [+6.4, +17.0]
Reversed yes/no report	+16.8 [+11.5, +22.2]	+24.0 [+18.5, +29.5]	+18.7 [+13.2, +24.3]
Admit what the passage omits	+11.8 [+6.4, +17.3]	+21.8 [+16.3, +27.2]	+15.4 [+10.0, +21.1]
Omission then report	+3.6 [−1.0, +8.6]	+14.6 [+9.4, +19.7]	+11.8 [+6.5, +17.2]
<i>Internal signals used directly</i>			
Output likelihood used directly	+5.8 [+0.6, +10.7]	+11.8 [+6.7, +16.9]	+10.2 [+5.3, +15.1]
Combined teacher used directly	+3.6 [−1.2, +8.2]	+8.8 [+3.9, +13.9]	+5.5 [+0.9, +10.0]

Table 9: Recall (%) at $k=2$, Qwen3-8B: development batteries (left), two frozen fresh Decoupled batteries (right). *: every seed exceeds the yes/no importance report ($p < 0.05$); Rating: 1–10 importance rating (Park et al., 2023).

Admission policy	Development				Fresh (held out)	
	Decoupled $n=68$	Compositional $n=52$	Confusable $n=68$	Unrelated $n=68$	Fresh-1 $n=316$	Fresh-2 $n=363$
Yes/no importance report	35.3	46.2	35.3	36.8	34.8	40.8
1–10 importance rating	47.1	53.8	48.5	48.5	43.7	45.7
SDA	49.5 ± 2.2	54.5 ± 1.1	48.5 ± 2.5	50.0 ± 2.5	44.2 ± 0.7*	50.0 ± 1.4*
Intended-bridge reference	51.5	65.4	51.5	51.5	50.0	54.3

Table 10: Recall (%) across models on the pool with unstated distractors (Fresh-2, 8.4 candidates per episode). p : largest p -value across seeds against each model’s own yes/no importance report across seeds.

Model	Yes/no report	SDA (3 seeds)	p vs. report	Intended-bridge reference
Qwen3-8B	41.0	51.0 / 51.2 / 49.9	$\leq 10^{-3}$	54.3
Qwen3-14B	32.2	40.8 / 39.4 / 39.9	$\leq 10^{-3}$	46.6
Qwen3-32B	41.0	51.0 / 50.7 / 50.1	$\leq 10^{-4}$	54.8
Qwen2.5-7B	23.7	31.4 / 28.4 / 30.6	$\leq .019$	31.7
OLMo-2-7B	38.8	52.3 / 51.0 / 51.2	$\leq 10^{-5}$	54.8
Mistral-7B	12.9	19.0 / 19.6 / 17.9	$\leq .015$	16.8

swering the question: for the primary SDA seed, recall is 54.6% with the bridge and 41.2% without it, versus 48.2% and 37.7% for the report.

The paired replacement experiment replays the primary SDA seed’s two selections. On the 249 episodes whose memory contains the bridge, we replace only that concept with SDA’s next-ranked candidate, keeping the other selection, question and frozen reader fixed. Recall changes from 54.6% to 37.8%, a 16.9-point decrease: 59 previously correct answers become incorrect and 17 become correct

(exact McNemar $p = 1.4 \times 10^{-6}$). Retaining the bridge supplies information used by the reader.

Bridge labels and reader success measure different outcomes. Qwen3-8B recall is 22.9% with the question alone, 63.9% with the whole passage, 54.3% with the intended-bridge reference and 50.0% with SDA. The intended-bridge reference prioritizes the intended bridge; the reader answers from the complete stored set and its own knowledge. On OLMo-2-7B and Mistral-7B, SDA

Table 11: Recall (%) on the enlarged pool (Fresh-2, 12.41 candidates per episode), Qwen3-8B. p : largest p -value across seeds against the yes/no importance report across seeds.

Budget	Yes/no report	Output likelihood	Workspace readout	SDA (3 seeds)	p vs. report	Intended-bridge reference
$k = 1$	29.5	28.1	24.0	38.0 / 38.0 / 39.7	$\leq 10^{-4}$	47.1
$k = 2$	40.5	32.0	34.7	50.7 / 52.6 / 51.2	$\leq 10^{-5}$	54.3
$k = 3$	41.9	33.9	38.6	52.9 / 52.1 / 52.1	$\leq 10^{-5}$	51.5

Table 12: Recall (%) across models on the held-out Fresh-2 battery, with k stored concepts. Rating: 1–10 importance rating (Park et al., 2023); *: every seed exceeds its model’s yes/no importance report at that budget ($p < 0.05$).

Model	$k=1$			$k=2$			$k=3$		
	Report	Rating	SDA	Report	Rating	SDA	Report	Rating	SDA
Qwen3-8B	28.1	34.7	37.6*	40.8	45.7	50.0*	41.6	47.9	52.2*
Qwen3-14B	23.4	31.1	33.5*	32.5	39.1	40.5*	37.5	43.5	43.6*
Qwen3-32B	32.5	32.0	41.4*	41.0	44.6	50.2*	46.0	50.7	53.0*
Qwen2.5-7B-Instruct	9.6	9.6	16.8*	23.1	22.6	29.8*	29.5	35.0	35.8*
OLMo-2-7B-Instruct	27.5	33.1	48.5*	38.0	44.1	55.4*	44.4	52.6	56.9*
Mistral-7B-Instruct	5.2	9.6	12.4*	12.7	15.7	19.8*	19.3	22.0	24.2*

Table 13: Admission policies on the standard pool and pool with unstated distractors, Qwen3-8B. Recall is in percent; rows above the rule are untrained baselines, and dashes denote unmeasured settings.

Admission policy	Standard pool			Pool with unstated distractors		
	$k = 1$	$k = 2$	$k = 3$	$k = 1$	$k = 2$	$k = 3$
Yes/no importance report	28.1	40.8	41.6	28.6	41.0	41.3
1–10 importance rating (Park et al., 2023)	34.7	45.7	47.9	35.8	46.6	48.8
Reversed yes/no report	35.5	46.6	49.0	23.4	28.9	30.9
Admit what the passage omits	46.6	53.4	49.6	31.7	35.3	36.9
Random selection	28.6	44.6	46.6	—	—	—
Output likelihood	—	48.8	—	30.6	39.1	40.5
Workspace readout	29.8	44.6	49.9	26.7	39.4	44.9
SDA (mean of 3 seeds)	37.6	50.0	52.2	37.6	50.7	51.6
Intended-bridge reference	47.1	54.3	51.5	47.1	54.3	51.5

reaches 55.4% and 19.8%, versus 54.8% and 16.8% for the intended-bridge reference.

the full-pool comparison measures the competition between stated and unstated candidates.

F Provenance, Wording and Pool-Composition Measurements

F.1 Ranking among candidates with the same provenance

The report discriminates usefulness among unstated candidates. Restricting the pool with unstated distractors to episodes with an unstated useful concept and at least one unstated negative leaves 345 episodes, with 5.01 qualifying candidates per episode on average. Within this group, report AUC is 0.880–0.897 across six models and workspace AUC is 0.592–0.697. Every paired 95% interval for workspace minus report lies below zero (Table 16). This measurement holds provenance fixed;

F.2 The full-pool AUC changes with the comparison pairs

On Fresh-2, the useful concept is stated in 5% of episodes and negative candidates in 99%; in 95% of episodes, the useful concept is the sole candidate absent from the passage. Adding unstated distractors leaves every original Qwen3-8B report score unchanged. It adds 1,450 negative candidates to the original 1,244, while retaining the same 363 positives. Useful-concept AUC against the added negatives is 0.889, compared with 0.322 against the original negatives. The pooled AUC is therefore

Table 14: Paired SDA minus baseline recall differences on three candidate pools, Qwen3-8B. Entries are in percentage points.

Baseline	$k = 1$	$k = 2$	$k = 3$
<i>Standard pool</i>			
Yes/no importance report	+9.6 [+5.8, +13.3]	+9.3 [+5.2, +13.2]	+10.6 [+6.8, +14.1]
1–10 importance rating	+2.9 [−0.6, +6.4]	+4.3 [+0.9, +7.8]	+4.2 [+1.3, +7.2]
Provenance-cued wording	+5.4 [+1.4, +9.4]	+7.1 [+2.9, +11.4]	+6.2 [+2.8, +9.7]
Reversed yes/no report	+2.1 [−3.3, +7.4]	+3.5 [−1.9, +8.9]	+3.1 [−1.3, +7.5]
Admit what the passage omits	−8.9 [−13.1, −4.6]	−3.4 [−7.7, +1.0]	+2.6 [−1.2, +6.2]
Omission then report	−9.5 [−13.7, −5.3]	−3.4 [−7.0, +0.4]	−0.5 [−3.2, +2.4]
Random selection	+9.0 [+4.1, +14.1]	+5.4 [+0.6, +10.3]	+5.6 [+1.4, +9.8]
<i>Pool with unstated distractors</i>			
Yes/no importance report	+8.9 [+5.1, +12.8]	+9.6 [+5.5, +13.8]	+10.3 [+6.3, +14.1]
1–10 importance rating	+1.7 [−1.6, +5.0]	+4.1 [+0.6, +7.6]	+2.9 [−0.4, +6.2]
Reversed yes/no report	+14.1 [+9.0, +19.5]	+21.8 [+16.2, +27.6]	+20.8 [+15.3, +26.1]
Admit what the passage omits	+5.9 [+0.6, +10.9]	+15.4 [+10.3, +20.8]	+14.7 [+9.6, +19.8]
Omission then report	−8.4 [−12.7, −4.2]	+5.8 [+0.8, +10.8]	+8.9 [+4.0, +13.8]
Output likelihood	+7.0 [+1.7, +12.0]	+11.6 [+6.5, +16.8]	+11.1 [+6.2, +16.1]
<i>Enlarged pool</i>			
Yes/no importance report	+9.1 [+5.2, +12.9]	+11.0 [+6.7, +15.2]	+10.5 [+6.6, +14.2]
1–10 importance rating	+2.2 [−1.3, +5.6]	+4.7 [+0.9, +8.4]	+4.1 [+0.9, +7.6]
Omission then report	−5.5 [−9.6, −1.5]	+6.6 [+1.6, +11.6]	+12.1 [+7.1, +17.2]
Output likelihood	+10.5 [+5.0, +15.7]	+19.6 [+14.3, +24.9]	+18.5 [+13.3, +23.6]
Random selection	+17.1 [+12.4, +22.0]	+24.5 [+19.1, +29.9]	+22.6 [+17.4, +27.7]

Table 15: Within-episode admission ranking on the standard Fresh-2 pool, Qwen3-8B. AUC differences are relative to the yes/no importance report; retention (%) measures whether the two admitted concepts include the useful concept.

Admission score	Per-episode AUC [95% CI]	Δ vs. report [95% CI]	Bridge retained ($k = 2$)
Yes/no report	0.326 [0.286, 0.366]	<i>reference</i>	29.8
1–10 rating (Park et al., 2023)	0.608 [0.567, 0.648]	+0.282 [+0.246, +0.320]	61.2
Workspace readout	0.547 [0.511, 0.584]	+0.221 [+0.172, +0.271]	51.0
Output likelihood	0.619 [0.582, 0.654]	+0.293 [+0.242, +0.344]	59.2
SDA (seed 0)	0.682 [0.641, 0.721]	+0.356 [+0.320, +0.393]	68.6
SDA (seed 1)	0.679 [0.639, 0.718]	+0.353 [+0.316, +0.391]	69.7
SDA (seed 2)	0.688 [0.648, 0.727]	+0.362 [+0.325, +0.401]	69.2

Table 16: Useful-concept ranking among unstated candidates, pool with unstated distractors. Qualifying episodes (345) average 5.01 candidates, including unstated useful and negative concepts. Entries: pooled AUC; $W - V$: workspace readout minus yes/no importance report.

Model	Yes/no report	Workspace readout	$W - V$ [95% CI]
Qwen3-8B	0.881	0.639	−0.242 [−0.274, −0.211]
Qwen3-14B	0.880	0.654	−0.226 [−0.257, −0.196]
Qwen3-32B	0.896	0.697	−0.199 [−0.228, −0.170]
Qwen2.5-7B	0.897	0.592	−0.305 [−0.340, −0.270]
OLMo-2-7B	0.883	0.661	−0.221 [−0.252, −0.191]
Mistral-7B	0.880	0.672	−0.208 [−0.238, −0.178]

the weighted average

$$\begin{aligned}
\text{AUC}_{\text{expanded}} &= \frac{1244}{2694} \text{AUC}_{\text{original}} \\
&\quad + \frac{1450}{2694} \text{AUC}_{\text{added}} \quad (9) \\
&= 0.627.
\end{aligned}$$

The added comparisons account for 53.8% of all positive–negative pairs. They raise AUC without changing the ordering of the original candidates. At a two-concept budget, the report continues to prioritize passage content: recall is 41.0%, against SDA’s 51.0%, 51.2% and 49.9% across seeds (Table 13).

F.3 Wording changes the importance ordering

Five matched wordings keep the episode, candidates and utility labels fixed. They ask about passage-only importance, remove that restriction, allow implied content (*implied-content wording*), state that the passage may omit the useful concept (*provenance-cued wording*), or specify a two-concept budget. The measurements cover four models on the standard pool and the pool with unstated distractors (Table 18). On Qwen3-32B, the provenance-cued wording raises standard-pool AUC from 0.398 to 0.876; the corresponding Qwen3-8B values are 0.321 and 0.537. On Mistral-7B, the implied-content wording gives 0.518, compared with 0.440 for passage-only importance. The direction and magnitude of the wording effect vary across models.

The provenance-cued wording’s ranking changes when the useful concept is stated. On Qwen3-8B, it reaches AUC 0.417 when the useful concept is named and 0.621 on Explicit. Its standard-pool recall is 32.2%, 43.0% and 46.0% at $k = 1, 2, 3$ (Table 14).

The strongest Qwen3-8B prompt baseline among wordings that permit inferences without specifying the useful concept’s provenance is the implied-content wording. Its recall rises from 28.1/40.8/41.6% to 29.2/41.3/44.9%. SDA trained and evaluated with this same wording reaches 37.1/50.7/51.0% (Table 17); all seeds exceed the wording alone at every budget, with exact McNemar $p \leq 6.9 \times 10^{-5}$. The passage-only wording SDA means are 37.6/50.0/52.2% (Table 13). Training through the implied-content wording adds recall beyond the wording change alone.

For wording j with prompt $u_j(c, x)$, the scorer is

$$s_j(c; x) = \log \sum_{t \in Y_{\text{word}}} p_{\mathcal{M}}(t \mid u_j(c, x)) - \log \sum_{t \in N_{\text{word}}} p_{\mathcal{M}}(t \mid u_j(c, x)). \quad (10)$$

The affirmative set contains the distinct token identities of *yes*, *Yes*, their leading-space forms, and *YES*, retaining a form only when the complete form is a single token; the negative set uses the analogous forms of *no*. The uppercase forms have no leading space in this scorer. All five wordings use these same sets, the answer-position logits, and thinking-disabled prompts. The main report instead uses the first token of each of six case-and-space forms,

as defined in App. B.6. The two aggregations account for the Mistral measurement difference in App. B.3.

Table 19 gives the exact question for each wording; it follows the passage, and *candidate* is replaced by the candidate concept.

F.4 Matched naming and position measurements

Naming the useful concept raises both internal readouts more than either importance report on Qwen3-8B, with episodes, candidates, labels and questions fixed. Workspace, output-likelihood, report and rating AUCs are initially 0.543, 0.613, 0.322 and 0.568. Appending a sentence naming the useful concept raises them to 0.952, 0.859, 0.515 and 0.581; placing the same sentence before the passage yields 0.916, 0.850, 0.582 and 0.655. Naming a distractor instead yields 0.393, 0.520, 0.249 and 0.519. Across six models, the stated-bridge control raises mean workspace AUC from 0.563 to 0.956 and mean report AUC from 0.373 to 0.504 (Table 20). The six individual report changes include Mistral-7B’s change from 0.411 to 0.395.

At $k = 2$, Qwen3-8B SDA recall on the stated-bridge control is 55.4%, 55.6% and 52.9%, against 44.9% for the report and 39.9% for the reversed report. Naming the bridge changes what an omission or reversal rule selects, while the trained module continues to retain useful information.

G Extended Ranking Measurements

G.1 Internal readouts and shuffled context

The report ranks useful content above the workspace readout on Explicit; the ordering reverses on Decoupled across all eight models (Table 22). Workspace rankings depend on the paired passage: shuffled context lowers workspace AUC across all six models while preserving candidates and labels (Table 21). All report measurements use Eq. (5) and the admission prompt in App. B.6.

Internal signals and the trained module use the passage. With shuffled context, workspace AUC falls across all six models, including 0.543 to 0.433 on Qwen3-8B and 0.581 to 0.423 on Mistral-7B (Table 21). Qwen3-8B likelihood AUC changes from 0.613 to 0.604 and report AUC from 0.322 to 0.485. Likelihood retains a context-independent candidate prior, while workspace ranking and the report’s preference for stated candidates depend on the paired passage. Replacing each passage

Table 17: Implied-content wording and distillation on the standard pool, Qwen3-8B. Recall is in percent; the final row gives paired recall differences in percentage points.

Admission policy	$k = 1$	$k = 2$	$k = 3$
Yes/no importance report	28.1	40.8	41.6
Implied-content wording	29.2	41.3	44.9
SDA with implied-content wording	37.1	50.7	51.0
SDA minus implied-content wording	+7.9 [+4.5, +11.4]	+9.4 [+5.8, +13.1]	+6.1 [+3.5, +8.9]

Table 18: Importance wording controls on Fresh-2 pools. Entries are pooled useful-concept AUCs from yes/no importance report log odds; all wordings and untrained models are shown.

Model	Importance wording	Standard pool	Pool with unstated distractors
Qwen3-8B	Passage only	0.321	0.628
	Unrestricted	0.391	0.663
	Implied-content wording	0.462	0.692
	Provenance-cued wording	0.537	0.726
	Two-concept budget	0.235	0.591
Qwen3-32B	Passage only	0.398	0.671
	Unrestricted	0.555	0.749
	Implied-content wording	0.656	0.802
	Provenance-cued wording	0.876	0.892
	Two-concept budget	0.327	0.633
OLMo-2-7B	Passage only	0.327	0.630
	Unrestricted	0.358	0.643
	Implied-content wording	0.420	0.675
	Provenance-cued wording	0.547	0.721
	Two-concept budget	0.301	0.614
Mistral-7B	Passage only	0.440	0.680
	Unrestricted	0.462	0.684
	Implied-content wording	0.518	0.716
	Provenance-cued wording	0.394	0.639
	Two-concept budget	0.336	0.626

Table 19: Five importance wordings, each following the passage.

Wording	Question
Passage only	Based only on the passage above, is the concept <i>candidate</i> one of the most important things to remember in order to answer possible future questions? Answer with a single word: yes or no.
Unrestricted	Is the concept <i>candidate</i> one of the most important things to remember in order to answer possible future questions about the passage above? Answer with a single word: yes or no.
Implied-content wording	Considering both what the passage above states and what it implies without stating, is the concept <i>candidate</i> one of the most important things to remember in order to answer possible future questions? Answer with a single word: yes or no.
Provenance-cued wording	Would storing the concept <i>candidate</i> help answer a likely future question about the passage above, even if the passage never mentions it? Answer with a single word: yes or no.
Two-concept budget	A memory system can keep only two concepts from the passage above. Should the concept <i>candidate</i> be one of them, so that a later question can be answered? Answer with a single word: yes or no.

with another episode’s also reduces SDA admission AUC from 0.682 to 0.526, 0.555 and 0.534 across seeds (within-episode AUC 0.540, 0.549 and 0.542), demonstrating a context-specific ranking.

G.2 Breadth across checkpoints, families and stages

Across the 23-checkpoint, four-benchmark grid in Table 22, workspace AUC exceeds report AUC on Decoupled and Compositional at every checkpoint with a defined report, including base and

Table 20: Matched stated-bridge control on Fresh-2, with the useful concept named. Entries: pooled AUC; W : workspace readout; LL: output likelihood; Report: yes/no importance report; Rating: 1–10 importance rating.

Model	W		LL		Report		Rating	
	Inferred	Stated	Inferred	Stated	Inferred	Stated	Inferred	Stated
Qwen3-8B	0.543	0.952	0.613	0.859	0.322	0.515	0.568	0.581
Qwen3-14B	0.571	0.954	0.636	0.864	0.355	0.607	0.565	0.591
Qwen3-32B	0.603	0.979	0.654	0.878	0.398	0.638	0.530	0.510
Qwen2.5-7B	0.520	0.879	0.600	0.890	0.427	0.456	0.481	0.481
OLMo-2-7B	0.561	0.986	0.633	0.911	0.327	0.410	0.533	0.585
Mistral-7B	0.581	0.990	0.630	0.854	0.411	0.395	0.608	0.601
Mean	0.563	0.956	0.628	0.876	0.373	0.504	0.548	0.558

Table 21: Candidate ranking with shuffled context on Fresh-2. Entries are pooled AUCs with the original or another episode’s passage; candidates and labels stay fixed.

Model	Workspace readout		Output likelihood		Yes/no report	
	Paired	Shuffled	Paired	Shuffled	Paired	Shuffled
Qwen3-8B	0.543	0.433	0.613	0.604	0.322	0.485
Qwen3-14B	0.571	0.473	0.636	0.625	0.355	0.404
Qwen3-32B	0.602	0.479	0.654	0.623	0.398	0.416
Qwen2.5-7B-Instruct	0.520	0.509	0.600	0.610	0.427	0.441
OLMo-2-7B-Instruct	0.561	0.477	0.633	0.617	0.327	0.412
Mistral-7B-Instruct	0.581	0.423	0.630	0.620	0.411	0.455

post-trained stages. An entry of “—” means that the tokenizer ships no chat template, so the yes/no importance report is undefined there.

Workspace AUC exceeds report AUC at every measured Qwen3 size on Decoupled, with non-monotonic paired changes between adjacent sizes (Table 23). From 8B to 32B, the paired change is -0.0536 (95% CI $[-0.1352, +0.0279]$).

Workspace AUC exceeds report AUC on Decoupled and Compositional at every released OLMo-2 post-training stage measured at 1B and 7B; the separation persists through SFT, DPO where available, and RLVR (Table 22).

G.3 Alternate-author and entity-bridge benchmarks

The report’s preference for stated candidates persists with a different author model. Across six models on the alternate-author benchmark, report AUC ranges from 0.220 to 0.384, alongside provenance AUC 0.650–0.803 (Table 24). These measurements use the same report definition as the primary batteries.

Report success accompanies a reversal of provenance preference. On the entity-bridge benchmark, Qwen2.5-7B and Qwen3-8B report AUCs reach 0.881 and 0.719, above workspace AUCs of 0.664 and 0.642 (Table 24). Both reports prefer

unstated candidates: their provenance AUCs are 0.133 and 0.289. Shuffled context reduces report AUC to 0.282 and 0.340. Qwen2.5-0.5B retains the usual preference for stated candidates, with provenance AUC 0.734 and utility AUC 0.270. The report reads the paired passage in both regimes; the direction of its provenance preference changes with candidate construction.

G.4 Admission after distillation across scales

SDA raises development admission AUC at every measured Qwen3 scale and seed (Table 25). All 18 paired 95% intervals for the gain lie above zero, while the largest absolute workspace AUC change is 0.0046. The measured improvement is concentrated in the admission interface.

Table 22: Predictive grid for 23 unique checkpoints across four benchmarks. Pooled item-level AUCs: W , workspace readout; V , yes/no importance report. Dashes: V undefined without a chat template; gray rows: base checkpoints; OLMo-2 Instruct: RLVR stage.

Checkpoint	Explicit		Implied-answer		Checkpoint	Decoupled		Compositional	
	W	V	W	V		W	V	W	V
Qwen2.5-0.5B	.5235	.4465	.5367	.4341	Qwen2.5-0.5B	.5019	.4798	.4293	.3715
Qwen2.5-0.5B-Instruct	.5211	.5496	.5158	.2916	Qwen2.5-0.5B-Instruct	.5261	.2841	.4190	.1776
Qwen2.5-1.5B	.5100	.5006	.5816	.4160	Qwen2.5-1.5B	.5903	.4052	.5451	.3879
Qwen2.5-1.5B-Instruct	.5139	.6136	.5769	.5234	Qwen2.5-1.5B-Instruct	.5768	.4637	.5489	.2936
Qwen2.5-3B	.5369	.5621	.6147	.4560	Qwen2.5-3B	.6062	.3784	.5208	.2444
Qwen2.5-3B-Instruct	.5331	.6446	.5781	.4615	Qwen2.5-3B-Instruct	.5951	.2586	.4971	.1951
Qwen2.5-7B	.5419	.5973	.6208	.4652	Qwen2.5-7B	.6121	.3730	.5152	.2771
Qwen2.5-7B-Instruct	.5376	.6792	.6341	.5856	Qwen2.5-7B-Instruct	.6423	.4897	.5156	.3472
Qwen3-0.6B	.4932	.5306	.5472	.3264	Qwen3-0.6B	.5364	.2248	.4693	.1123
Qwen3-1.7B	.4818	.5917	.5409	.3895	Qwen3-1.7B	.5607	.1961	.4162	.1812
Qwen3-4B	.5090	.6496	.5744	.4798	Qwen3-4B	.5291	.3967	.5050	.2600
Qwen3-8B	.5090	.7100	.6204	.4534	Qwen3-8B	.6545	.3872	.5466	.2079
Qwen3-14B	.5083	.7400	.6390	.5219	Qwen3-14B	.6677	.4271	.5086	.2163
Qwen3-32B	.5084	.7480	.6139	.5511	Qwen3-32B	.6919	.4782	.5532	.2774
OLMo-2-1B	.5006	—	.6308	—	OLMo-2-1B	.6001	—	.5114	—
OLMo-2-1B-SFT	.5409	.5252	.5969	.5016	OLMo-2-1B-SFT	.5725	.3716	.4760	.3064
OLMo-2-1B-DPO	.5374	.5563	.5759	.4973	OLMo-2-1B-DPO	.5802	.3799	.4764	.2786
OLMo-2-1B-Instruct	.5354	.5551	.5795	.4899	OLMo-2-1B-Instruct	.5801	.3284	.4783	.2548
OLMo-2-7B	.4806	—	.6735	—	OLMo-2-7B	.6835	—	.5453	—
OLMo-2-7B-SFT	.4959	.5827	.6827	.5369	OLMo-2-7B-SFT	.6637	.3317	.5312	.2924
OLMo-2-7B-Instruct	.4938	.6231	.6688	.5398	OLMo-2-7B-Instruct	.6664	.3529	.5341	.2944
Mistral-7B-Instruct-v0.3	.5324	.6191	.6687	.5777	Mistral-7B-Instruct-v0.3	.6713	.4700	.5962	.3242
GPT-2	.5070	—	.5388	—	GPT-2	.5331	—	.5170	—

Table 23: Qwen3 dense ladder on Decoupled. Upper block: workspace readout minus yes/no importance report ($W - V$); lower block: changes in $W - V$ between sizes. *: 95% interval excludes zero.

Checkpoint	$W - V$ [95% CI]
Qwen3-1.7B	+0.3646* [+0.2709, +0.4522]
Qwen3-4B	+0.1324* [+0.0280, +0.2346]
Qwen3-8B	+0.2673* [+0.1670, +0.3668]
Qwen3-14B	+0.2405* [+0.1392, +0.3409]
Qwen3-32B	+0.2137* [+0.1119, +0.3177]
Size change	$\Delta(W - V)$ [95% CI]
1.7B $\rightarrow$ 4B	-0.2322* [-0.3338, -0.1321]
4B $\rightarrow$ 8B	+0.1349* [+0.0521, +0.2147]
8B $\rightarrow$ 14B	-0.0268 [-0.0949, +0.0475]
14B $\rightarrow$ 32B	-0.0268 [-0.1057, +0.0536]
8B $\rightarrow$ 32B	-0.0536 [-0.1352, +0.0279]

Table 24: Replications on the alternate-author benchmark and entity-bridge benchmark. Pooled AUCs: W , workspace readout; V , thinking-disabled yes/no importance report; $W - V$, their difference. *: 95% interval excludes zero.

Benchmark	Model	W	V	$W - V$ [95% CI]
Alternate-author	Qwen2.5-7B	0.490	0.384	+0.106 [-0.009, +0.215]
	Qwen3-8B	0.515	0.220	+0.295 [+0.197, +0.386]*
	Qwen3-14B	0.521	0.267	+0.254 [+0.166, +0.340]*
	Qwen3-32B	0.522	0.275	+0.247 [+0.134, +0.356]*
	OLMo-2-7B	0.547	0.353	+0.194 [+0.093, +0.297]*
	Mistral-7B	0.507	0.294	+0.213 [+0.128, +0.300]*
Entity-bridge	Qwen2.5-0.5B	0.569	0.270	+0.299 [+0.211, +0.396]*
	Qwen2.5-7B	0.664	0.881	-0.217 [-0.303, -0.133]*
	Qwen3-8B	0.642	0.719	-0.076 [-0.189, +0.032]

Table 25: Admission ranking on the Decoupled and Compositional development batteries across Qwen3 scales. V : pooled admission AUC; ΔV : paired change; $|\Delta W|$: largest absolute workspace readout change across seeds.

Model	Benchmark	V before	Seed	V after	ΔV [95% CI]	$ \Delta W $
Qwen3-8B	Decoupled	0.387	0	0.756	+0.369 [+0.310, +0.431]	0.0033
			1	0.739	+0.352 [+0.293, +0.417]	
			2	0.758	+0.371 [+0.312, +0.435]	
Qwen3-8B	Compositional	0.208	0	0.618	+0.410 [+0.325, +0.490]	0.0021
			1	0.623	+0.415 [+0.329, +0.494]	
			2	0.619	+0.412 [+0.326, +0.491]	
Qwen3-14B	Decoupled	0.427	0	0.702	+0.275 [+0.216, +0.340]	0.0029
			1	0.669	+0.242 [+0.192, +0.299]	
			2	0.680	+0.253 [+0.200, +0.312]	
Qwen3-14B	Compositional	0.216	0	0.529	+0.312 [+0.234, +0.395]	0.0029
			1	0.458	+0.242 [+0.172, +0.317]	
			2	0.481	+0.265 [+0.190, +0.346]	
Qwen3-32B	Decoupled	0.478	0	0.773	+0.295 [+0.237, +0.356]	0.0037
			1	0.762	+0.284 [+0.224, +0.346]	
			2	0.721	+0.243 [+0.186, +0.299]	
Qwen3-32B	Compositional	0.277	0	0.622	+0.344 [+0.273, +0.418]	0.0046
			1	0.624	+0.347 [+0.267, +0.427]	
			2	0.588	+0.310 [+0.233, +0.388]	

**H Teacher and Optimization
Measurements**

**H.1 Teacher composition and candidate
readout**

Table 26 reports primary-pool recall for all three teacher compositions; Table 27 gives their complete paired intervals.

Table 26: Teacher composition on the statedness-balanced pool, five models. Recall (%); reduced columns show students trained with one teacher signal removed. Table 27 gives paired differences.

Model	Budget	SDA	Without the workspace readout	Without the output likelihood
Qwen3-8B	$k = 1$	40.5	42.1	36.7
	$k = 2$	53.7	54.8	49.8
	$k = 3$	51.8	50.6	50.5
Qwen3-14B	$k = 1$	35.5	36.2	34.3
	$k = 2$	44.1	43.4	42.4
	$k = 3$	45.3	45.8	45.2
Qwen3-32B	$k = 1$	44.9	43.4	45.7
	$k = 2$	53.3	52.2	54.7
	$k = 3$	54.8	55.3	56.6
Qwen2.5-7B	$k = 1$	17.4	16.2	17.6
	$k = 2$	33.3	32.2	32.4
	$k = 3$	39.1	37.6	37.3
OLMo-2-7B	$k = 1$	49.5	46.2	45.2
	$k = 2$	54.7	53.4	51.1
	$k = 3$	55.5	55.5	55.2

Table 27: Teacher signal ablations on the statedness-balanced pool, five models. Entries are SDA minus students trained with one teacher signal removed, in recall percentage points.

Model	Teacher	$k = 1$	$k = 2$	$k = 3$
Qwen3-8B	Without workspace readout	-1.6 [-4.0, +0.5]	-1.1 [-3.7, +1.6]	+1.2 [-1.2, +3.6]
	Without output likelihood	+3.8 [+1.6, +6.2]	+4.0 [+1.9, +6.1]	+1.3 [-0.6, +3.3]
Qwen3-14B	Without workspace readout	-0.7 [-2.7, +1.2]	+0.6 [-0.6, +2.0]	-0.6 [-2.0, +0.9]
	Without output likelihood	+1.1 [-0.2, +2.5]	+1.6 [+0.1, +3.4]	+0.1 [-1.4, +1.6]
Qwen3-32B	Without workspace readout	+1.5 [+0.1, +2.9]	+1.0 [-0.1, +2.2]	-0.5 [-1.8, +1.0]
	Without output likelihood	-0.8 [-2.9, +1.1]	-1.5 [-3.1, +0.2]	-1.7 [-3.6, +0.0]
Qwen2.5-7B	Without workspace readout	+1.1 [-0.5, +2.6]	+1.1 [-0.6, +2.9]	+1.5 [+0.0, +2.9]
	Without output likelihood	-0.3 [-1.6, +1.1]	+0.9 [-1.1, +2.9]	+1.8 [+0.3, +3.5]
OLMo-2-7B	Without workspace readout	+3.3 [+1.2, +5.5]	+1.3 [-0.5, +3.2]	+0.0 [-1.9, +1.8]
	Without output likelihood	+4.3 [+2.0, +6.7]	+3.6 [+1.2, +6.1]	+0.3 [-1.9, +2.5]

We construct the paired-passage battery after the Fresh-2 freeze from the 363 frozen Fresh-2 episodes. We match these episodes into 181 disjoint pairs, yielding 362 episodes. Each source episode appears in at most one pair, and one episode remains unpaired. Each member retains its own passage, future question, and answer. Its own bridge is its only useful candidate, and its partner’s bridge is a negative candidate. Each member’s candidate set is the union of the two members’ original candidate lists. The resulting sets contain 8.86 candidates per episode on average, with between 8 and 12 candidates. Each member presents the union in its own shuffled order. No member retains its source order, and the two members of a pair never share an order. Both bridges are absent from the passage in all 362 episodes. Statedness cannot separate the useful candidate from the partner’s bridge; only the content of the passage can. SDA’s configuration is fixed before Fresh-2 (App. C), so no

configuration choice uses this battery. Figure 6 shows one complete pair. We evaluate Qwen3-8B on all 362 episodes. The per-model and pooled analyses use the same pairs.

Table 28 gives the paired-passage results for each model.

Table 29 reports SDA recall of 38.1/49.1/52.3% at one, two and three stored concepts, with 48.2% retention at one. Removing the workspace readout from SDA’s teacher costs 5.3 retention points ([+2.4, +8.4]) and leaves recall unchanged on this set: the difference is -0.1 points ([-2.0, +1.8]). Removing the output likelihood costs both, 2.9 retention points ([+0.1, +5.7]) and 3.0 recall points ([+1.1, +4.9]). These differences are seed-averaged, with 95% bootstrap intervals over episodes from 4,000 draws. Removing the workspace readout also costs 3.1 and 3.5 retention points at two and three stored concepts, with both intervals excluding zero. The both-of-a-pair

Example 3: Paired passage, first member

Passage. Soraya carefully arranged the menorah on the windowsill, lighting the first candle at sunset. Her family gathered, singing traditional songs and spinning dreidels. Plates of latkes and sufganiyot were passed around. The evening was filled with laughter and storytelling, each tale passed down for generations.

Candidates, in supplied order. clients, bouquets, lilies, florist, hanukkah, song, latke, candle, holiday, vases.
Useful concept. hanukkah.

Future question. What religion does Soraya traditionally observe during this holiday?

Answer. judaism.

Example 4: Paired passage, second member

Passage. Minh spent the afternoon arranging yellow and white lilies in tall crystal vases. She snipped stems and removed wilted leaves to ensure the displays would last. After consulting her chart, she delivered the finished bouquets to waiting clients preparing for a special event downtown.

Candidates, in supplied order. florist, latke, song, holiday, clients, vases, candle, bouquets, hanukkah, lilies.

Useful concept. florist.

Future question. What might Minh worry about if her workplace got too cold overnight?

Answer. freezing.

Figure 6: One pair from the paired-passage battery. Both passages leave *hanukkah* and *florist* unstated and share ten candidates in different orders; the passage determines which bridge is useful.

Table 28: Paired-passage battery results by model. Entries are SDA minus students trained with one teacher signal removed, in percentage points; retention indicates admission of the episode’s useful concept.

Model	Budget	Without the workspace readout		Without the output likelihood	
		Retention	Recall	Retention	Recall
Qwen3-8B	$k = 1$	+5.3 [+2.5, +8.2]	−0.1 [−2.0, +1.8]	+2.9 [+0.0, +5.8]	+3.0 [+1.1, +4.8]
	$k = 2$	+3.1 [+0.2, +6.2]	−0.2 [−2.6, +2.1]	+2.4 [−0.2, +5.2]	+3.2 [+1.0, +5.5]
	$k = 3$	+3.5 [+1.1, +6.0]	+1.7 [−0.6, +3.8]	+2.2 [+0.1, +4.3]	+2.3 [+0.3, +4.3]
Qwen3-14B	$k = 1$	+9.0 [+6.5, +11.7]	+1.4 [−0.6, +3.5]	−1.4 [−3.5, +0.7]	+0.3 [−1.2, +1.8]
	$k = 2$	+6.0 [+3.8, +8.5]	+0.0 [−1.7, +1.8]	−1.3 [−3.3, +0.7]	+1.3 [−0.5, +3.1]
	$k = 3$	+2.3 [+0.6, +4.0]	+0.6 [−0.9, +2.0]	+1.6 [−0.1, +3.3]	+1.8 [+0.0, +3.5]
Qwen3-32B	$k = 1$	+3.8 [+2.1, +5.6]	+1.6 [+0.2, +3.0]	−10.6 [−13.3, −7.9]	−1.9 [−4.0, +0.1]
	$k = 2$	+1.8 [−0.1, +3.7]	+0.7 [−0.6, +2.0]	−6.2 [−8.0, −4.4]	−0.3 [−2.2, +1.7]
	$k = 3$	+0.7 [−0.6, +2.2]	+0.2 [−0.9, +1.2]	−4.5 [−6.3, −2.9]	−0.8 [−2.3, +0.6]
Qwen2.5-7B	$k = 1$	+8.3 [+6.3, +10.3]	+3.0 [+1.7, +4.5]	−1.6 [−4.0, +0.8]	+0.8 [−0.7, +2.5]
	$k = 2$	+6.2 [+4.5, +7.9]	+1.8 [+0.5, +3.2]	+0.6 [−1.2, +2.4]	+0.6 [−1.6, +2.6]
	$k = 3$	+4.4 [+2.8, +6.2]	+1.5 [−0.1, +3.1]	+4.3 [+2.6, +6.2]	+0.4 [−1.5, +2.1]
OLMo-2-7B	$k = 1$	+10.0 [+7.6, +12.5]	+3.4 [+1.5, +5.4]	+3.9 [+0.8, +6.8]	+4.4 [+2.4, +6.5]
	$k = 2$	+6.0 [+4.1, +7.9]	+0.7 [−1.0, +2.4]	+5.0 [+2.5, +7.5]	+3.1 [+0.6, +5.6]
	$k = 3$	+4.0 [+2.5, +5.5]	+1.7 [+0.5, +2.9]	+4.8 [+2.6, +7.1]	+3.4 [+1.3, +5.5]

column reports 20.8% for SDA, 17.9% for its teacher without the workspace readout and 20.4% for its teacher without the output likelihood. A passage-blind candidate ordering cannot raise this rate above zero at one stored concept. Against the untrained baselines at one stored concept, SDA improves recall by 10.5 points over the yes/no importance report ([+6.7, +14.3]) and by 2.5 points over the 1–10 importance rating ([−1.0, +6.0]).

Table 30 gives the replaced-passage measurements of Figure 4 together with the untrained workspace readout and yes/no report.

Over the 220 teacher-labeled episodes, each containing a useful candidate, SDA’s teacher’s two positive labels differ from those of the teacher with-

out the workspace readout on 43.2% of episodes and from those of the teacher without the output likelihood on 45.0%. On disagreement episodes, SDA’s teacher’s labels contain the useful concept in 63.2% against 58.9% for the teacher without the workspace readout (exact McNemar $p = 0.67$), and in 63.6% against 48.5% for the teacher without the output likelihood ($p = 0.032$). Across all 220 episodes, these rates are 60.9%, 59.1% and 54.1%, respectively.

The teacher uses each candidate’s first token as specified in Eq. (3). The full-candidate readout protocol appends the candidate and reads each token at the position preceding it. Each token receives its peak reciprocal rank over layers; the candidate

Table 29: Paired-passage battery results, Qwen3-8B. Recall and useful-concept retention (%); final column: pairs where both members retain their own useful concept (%). Rows naming signals: students of teachers with that signal removed.

Admission policy	Recall			Retention	
	$k = 1$	$k = 2$	$k = 3$	$k = 1$	Both of a pair
SDA	38.1	49.1	52.3	48.2	20.8
Without the workspace readout	38.2	49.3	50.6	42.8	17.9
Without the output likelihood	35.2	45.9	50.0	45.3	20.4
1–10 importance rating (Park et al., 2023)	35.6	43.6	46.1	39.5	13.8
Yes/no importance report	27.6	38.7	43.4	18.8	3.9
Output likelihood used directly	26.8	39.5	40.6	21.6	3.3
Workspace readout used directly	22.6	34.0	37.3	17.1	5.0

Table 30: Passage replacement on the standard Fresh-2 pool, Qwen3-8B. Retention (%) measures whether the single stored concept is useful; drops are own-passage retention minus retention with another episode’s passage.

Admission policy	Own passage	Replaced passage	Drop
SDA	48.3	27.4	+20.9 [+14.3, +27.4]
Without the workspace readout	42.8	39.9	+2.9 [−3.5, +9.2]
Without the output likelihood	46.6	20.1	+26.5 [+20.5, +32.6]
Workspace readout, untrained	26.2	14.3	+11.8 [+7.4, +16.2]
Yes/no report, untrained	16.0	20.4	−4.4 [−9.9, +1.1]

score aggregates these peaks by their mean or minimum. Each aggregation and the first-token readout take their own maximum over surface forms.

On the standard Fresh-2 pool with Qwen3-8B, 56% of candidates span multiple tokens. Averaging over all candidate tokens improves ranking over the matched first-token readout (Table 31).

Teacher composition on the standard pool. On the standard Fresh-2 pool (Table 32), the teacher without the workspace readout reaches 50.3%, a paired +0.3 percentage points relative to SDA (95% CI [−2.2, +2.8]); the teacher without the output likelihood reaches 47.9%, a paired −2.1 points ([−4.1, −0.3]).

Distillation changes the usable ranking. Using the combined teacher directly yields 45.7% standard-pool Fresh-2 recall, −4.3 points relative to SDA ([−8.9, +0.1]). Direct workspace scoring yields 44.6%, a difference of −5.4 points ([−10.4, −0.5]), and direct likelihood yields 48.8%, a difference of −1.3 points ([−5.6, +2.8]). The student learns a context-and-candidate judgment from the teacher’s ordering.

H.2 The training objective determines admission ordering

The objective baseline aligns training with the admission score by applying binary cross-entropy

to the yes/no log-odds in Eq. (5). Teacher labels, training episodes, adapter shape and the 414-step budget are fixed. Validation within-episode admission AUC starts at 0.657 for the untrained model and reaches 0.824 at the final step under SDA’s vocabulary-normalized token objective. At learning rate 10^{-5} , the log-odds objective ends at 0.250, 0.333 and 0.301 across the three seeds. Validation selects the untrained checkpoint in all three runs (Table 33).

At a step size five times smaller (2×10^{-6}), validation selects a trained checkpoint after 100 steps, with within-episode AUC 0.690 and standard-pool recall of 40.2% at $k = 2$, against 50.0% for SDA’s vocabulary-normalized token prediction. At 5×10^{-6} , selection returns the untrained checkpoint. Each smaller-step-size run uses one seed. SDA’s token objective yields 9.8 points more recall than the trained log-odds checkpoint and 9.6 points more than the untrained checkpoint selected at the other learning rates.

Reader reward adds task-specific supervision. SDA exceeds recall-reward training by 3.7 points on Fresh-2 ([+1.3, +6.2]); recall-reward refinement after SDA adds 2.8 points ([+0.6, +5.1]). These ablations use the same training episodes and adapter shape with a matched step budget (App. B.7).

Table 31: Candidate readout comparison on the standard pool, Qwen3-8B, with 1,607 candidates. Entries are pooled and within-episode AUCs.

Workspace readout	Pooled AUC	Within-episode AUC
First token	0.530	0.536
Mean across all tokens	0.574	0.571
Minimum across all tokens	0.531	0.538

Table 32: SDA ablations on the standard Fresh-2 pool, Qwen3-8B, $k = 2$. Recall is in percent; Δ is the recall difference from SDA in percentage points.

Variant	Recall	Δ vs. SDA
SDA	50.0	<i>reference</i>
<i>Teacher with one signal removed</i>		
Without the output likelihood	47.9	−2.1 [−4.1, −0.3]
Without the workspace readout	50.3	+0.3 [−2.2, +2.8]
<i>No distillation</i>		
Combined teacher used directly	45.7	−4.3 [−8.9, +0.1]
<i>Recall reward</i>		
Recall reward instead of distillation	46.4	−3.7 [−6.2, −1.3]
SDA followed by recall-reward refinement	52.9	+2.8 [+0.6, +5.1]

Table 33: Objective ablation on the standard pool, Qwen3-8B, $k = 2$. Recall is in percent; main rows use learning rate 10^{-5} , and each smaller-rate row uses one seed.

Training objective	Recall
SDA: token-level cross-entropy	50.0
Yes/no log-odds cross-entropy	40.5
Yes/no log-odds, learning rate 2×10^{-6}	40.2
Yes/no log-odds, learning rate 5×10^{-6}	40.5

I Prompts

Every natural-language instruction used in the paper is reproduced below. *Italic phrases* stand for the supplied passage, candidate, memory list or question.

Prompt 1: Episode authoring

Design a batch of diverse test episodes about silent bridge reasoning for a memory experiment. Each episode has a passage of four to eight sentences about a named person; invent a distinctive first name, different in every episode, and vary the domain widely. A question asked later, out of context, refers to the episode only through the person's name and requires an unstated bridge fact inferable from the passage, followed by common general knowledge. Supply a short correct answer of one to three words and five to seven candidate concepts, each a common lowercase word with a utility category and a short reason.

Exactly one candidate is the useful bridge concept, and its word must be absent from the passage. The question must share no passage content word except the person's name: someone who never read the passage must be unable to answer it. For a passage about Elena wandering the Plaza Mayor, ask what language Elena should have brushed up on for her trip, without naming the plaza in the question. The answer must differ from every candidate and be absent from the passage; combining the bridge with general knowledge yields the answer. For example, the Plaza Mayor cues *madrid*, which supports *spanish*. A reader remembering only the bridge plus the question must be able to answer.

Include two or three vivid, emotional or surprising distractors that occur literally in the passage but are useless for the question, and one or two neutral concepts that occur in the passage. Every candidate must be a single common English lowercase word. Do not reuse concepts or person names across episodes.

Prompt 2: Admission, shared by SDA and the report

Passage

Based only on the passage above, is the concept "*candidate concept*" one of the most important things to remember in order to answer possible future questions? Answer with a single word: yes or no.

Prompt 3: 1–10 importance rating

Passage

On a scale of 1 to 10, where 1 is purely mundane and 10 is extremely important, rate how important it is to remember the concept "*candidate concept*" in order to answer possible future questions about this passage. Answer with a single number.

Prompt 4: ConsistencyGate, sampled form

You are a fact-checking assistant. Given a source context and a candidate fact, rate how strongly the fact is supported by the context.

Source context: *passage*

Candidate fact: *candidate concept*

Question: How strongly is this candidate fact supported by the source context above? Do not use outside knowledge.

0.0 = completely unsupported or directly contradicted by the context

0.5 = partially supported or ambiguous

1.0 = fully and directly supported by the context

Output only a single decimal number between 0.0 and 1.0.

Prompt 5: ConsistencyGate, single-pass form

You are a fact-checking assistant. Given a source context and a candidate fact, decide whether the fact is supported by the context.

Source context: *passage*

Candidate fact: *candidate concept*

Question: Is this candidate fact supported by the source context above? Do not use outside knowledge.

Answer with a single word: yes or no.

Prompt 6: A-MAC utility

passage

Consider storing the concept “*candidate concept*” in long-term memory. Is it actionable, does it support likely follow-up questions, or does it capture a persistent constraint or preference?

Answer with a single word: yes or no.

Prompt 7: Context-only candidate proposal

Passage

A memory system can store only a few single-word concepts about the passage above. List up to 8 concepts that could be worth remembering for possible future questions, including things the passage clearly implies but does not state. Answer with lowercase single words separated by commas, and nothing else.

Prompt 8: Frozen reader

Earlier you read a passage. The only things you remember from it are: *stored concepts separated by commas*.

Using only what you remember, answer concisely.

Question: *future question*

Answer:

Prompt 9: Blind semantic judge

Grade short answers to memory questions. For each item decide whether ANSWER gives GOLD as its answer.

Labels:

“correct”: the answer commits to GOLD or an exact equivalent (synonym, inflection, fuller name) as its single answer.

Extra explanation is fine.

“hedged”: it offers two or more alternative answers (e.g. “X or Y”), or states the answer cannot be determined, even if GOLD is mentioned.

“incorrect”: anything else, including a different answer.